

Naučni metod

od slučajnosti do teorije

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Da li je quicksort brz?

Kako to znamo?

Plan

1. Naučni metod
2. Jezik slučajnosti
3. Merenja i testovi
4. Slučajni algoritmi
5. Istraživanje: **Quicksort**
6. Istraživanje: **slučajni grafovi**
7. Od rezultata do teorije

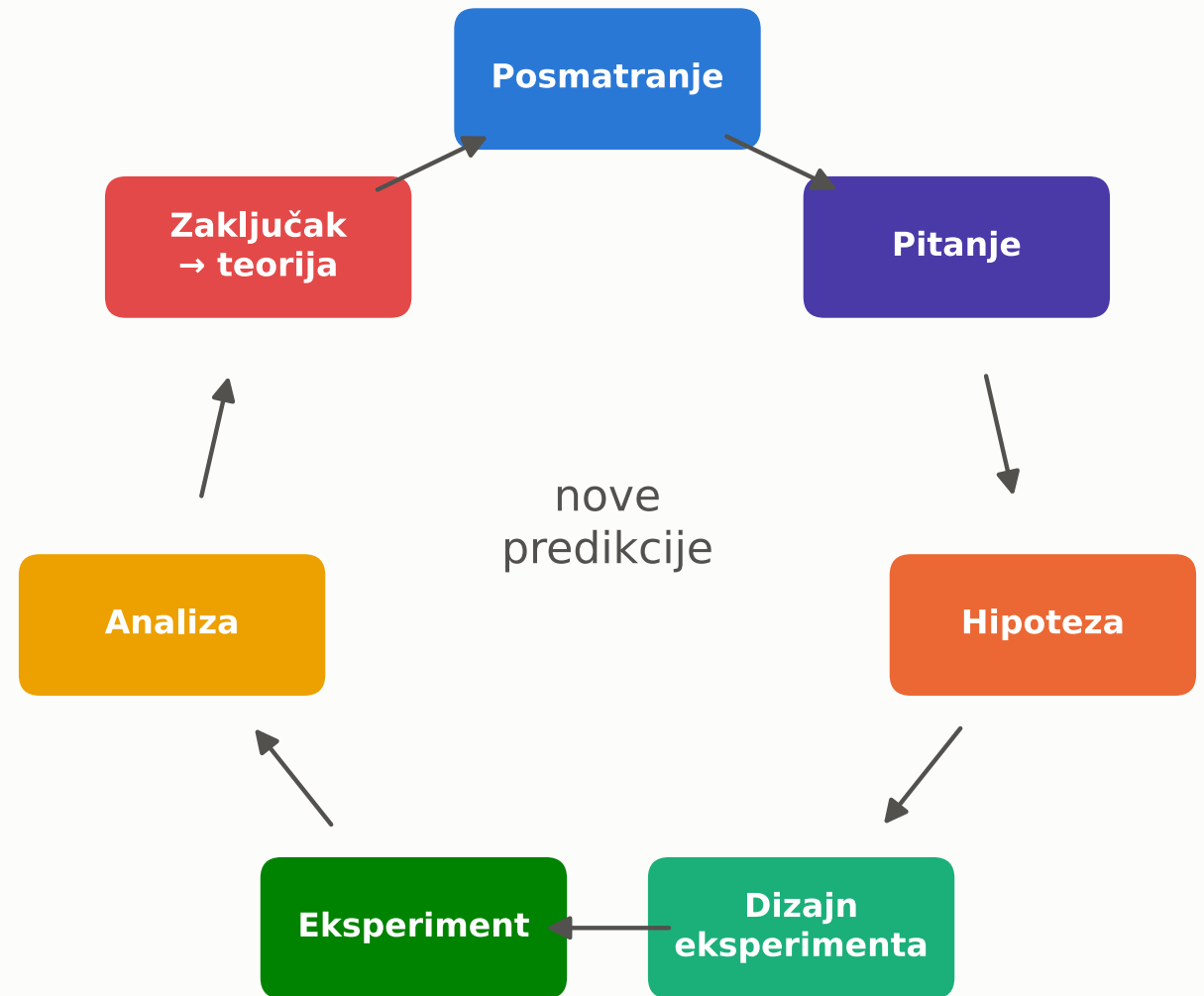
I

Nauční metod

Ciklus

- posmatramo
- pitamo
- predviđamo
- proveravamo
- zaključujemo

Svaki krug daje **nova pitanja**.



Posmatranje → pitanje

„Quicksort je nekad brz, a nekad spor.”

Koliko poređenja C_n pravi quicksort nad n elemenata?

Dobro pitanje je **merljivo**.

Hipoteza

Tvrdnja koja **može da bude netačna**.

Loše	Dobro
„Quicksort je dobar.”	$E[C_n] \leq 2n \ln n$
„Grafovi su povezani.”	$np > 1 \Rightarrow$ postoji džinovska komponenta
„Lek pomaže.”	$\mu_{\text{lek}} < \mu_{\text{placebo}}$

Falsifikabilnost

$$H : \quad \forall x \left(\text{labud}(x) \Rightarrow \text{beo}(x) \right)$$

Hiljadu belih labudova **ne dokazuje** H .

$$\exists x \left(\text{labud}(x) \wedge \neg \text{beo}(x) \right) \Rightarrow \neg H$$

Jedan crni labud je **obara**.

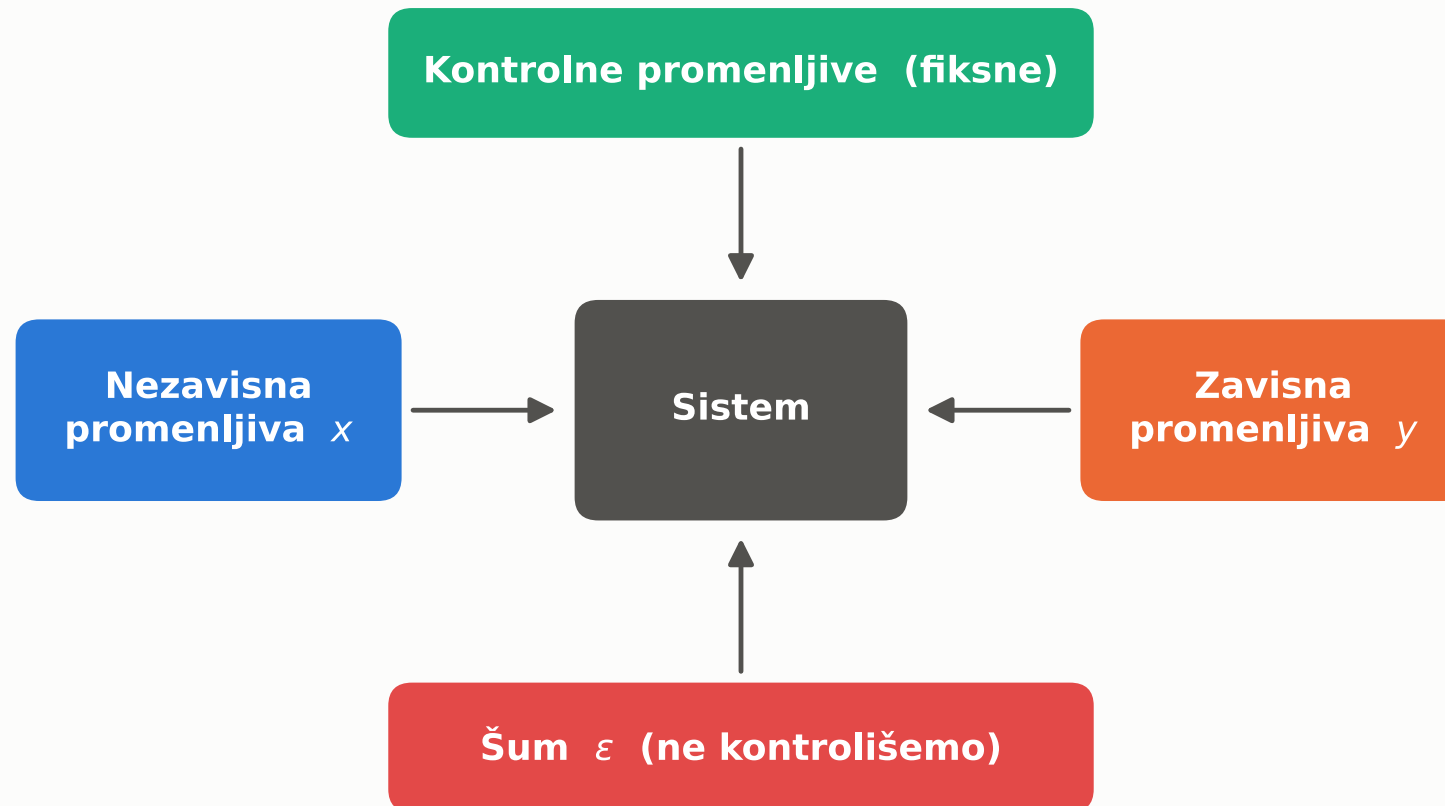
Nulta i alternativna hipoteza

$$H_0 : \mu_A = \mu_B \quad \text{„nema efekta”}$$

$$H_1 : \mu_A \neq \mu_B \quad \text{„ima efekta”}$$

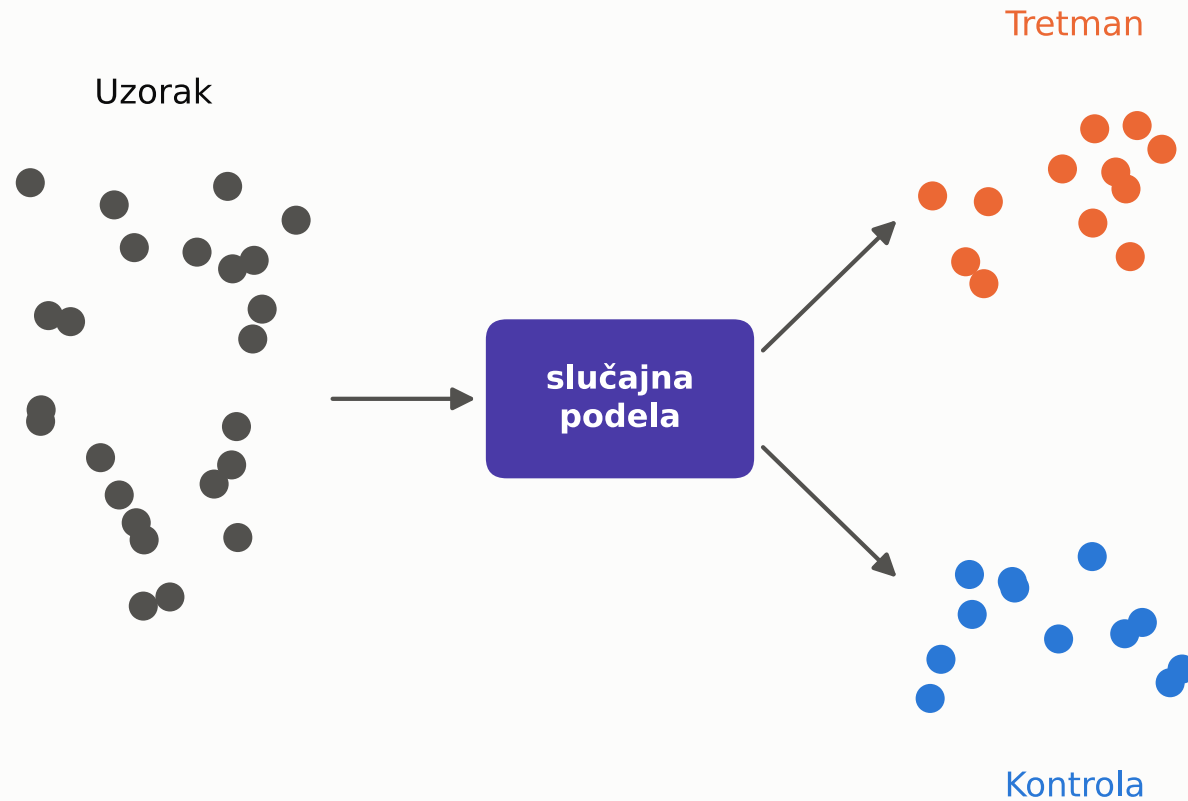
Pokušavamo da **odbacimo** H_0 .

Eksperimentalni dizajn



$$y = f(x) + \varepsilon$$

Kontrola i randomizacija



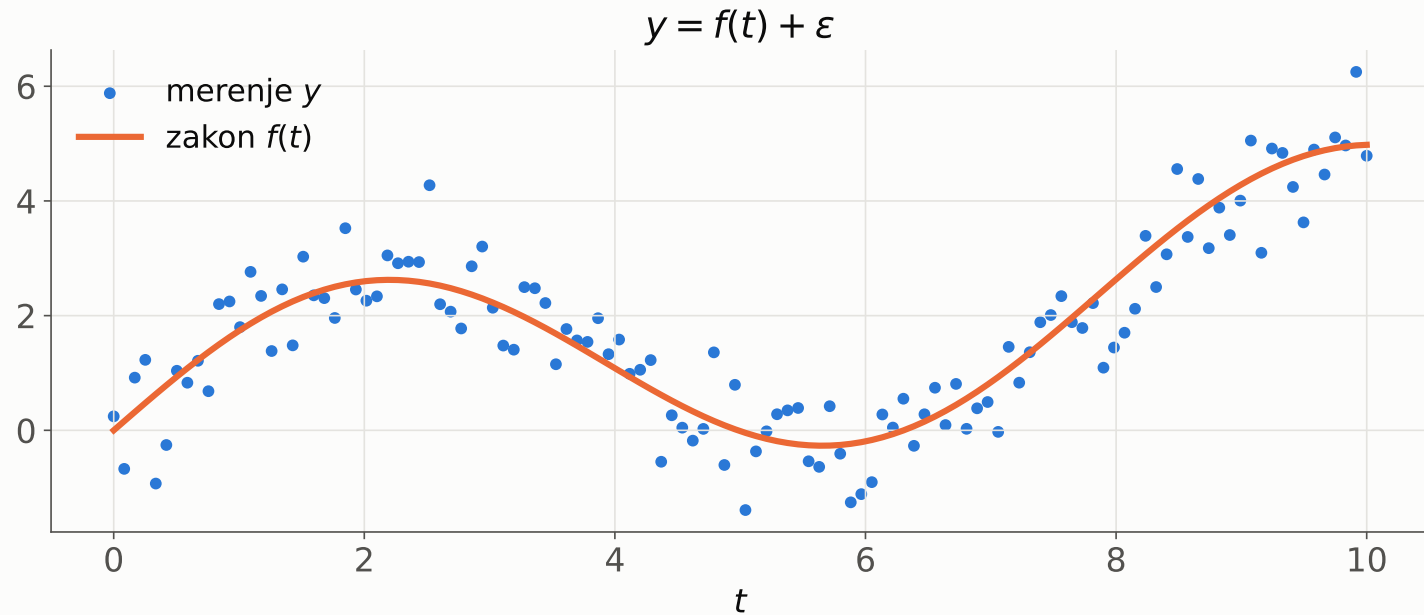
Kontrola	Randomizacija	Ponavljanje

Eksperiment

Merenje nikad nije savršeno.

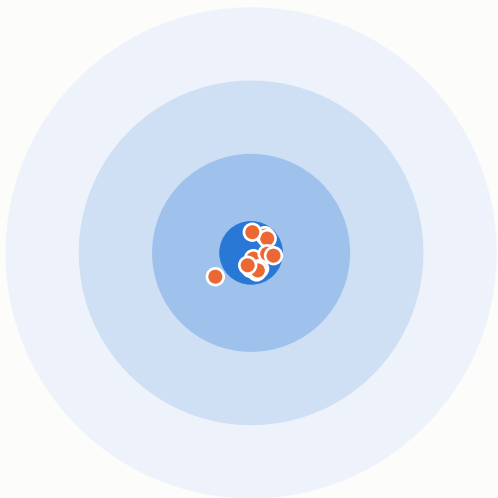
$$y = f(x) + \varepsilon$$

- f — zakon koji tražimo
- ε — šum

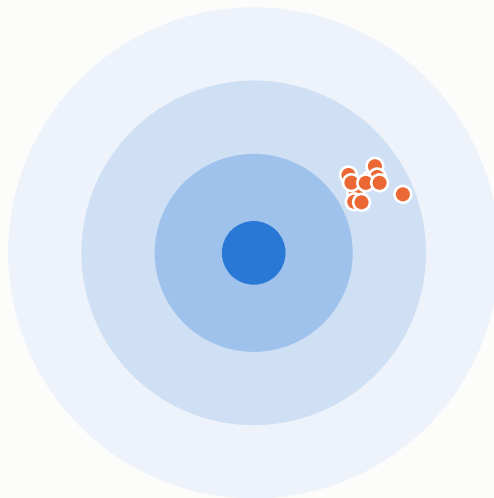


Tačnost i preciznost

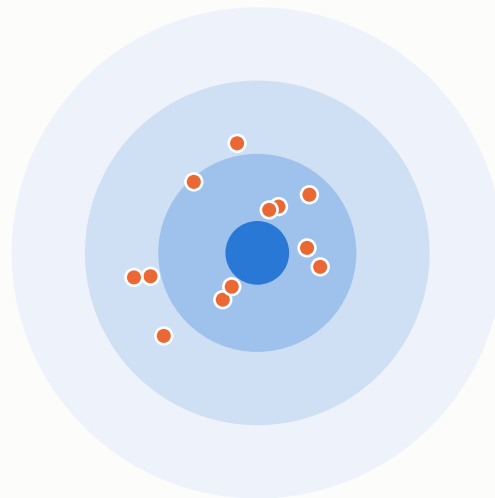
tačno i precizno



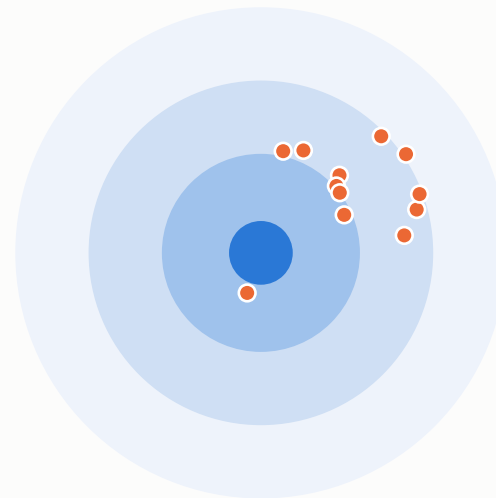
netačno, precizno



tačno, neprecizno



netačno i neprecizno



$$y = f(x) + b + \varepsilon$$

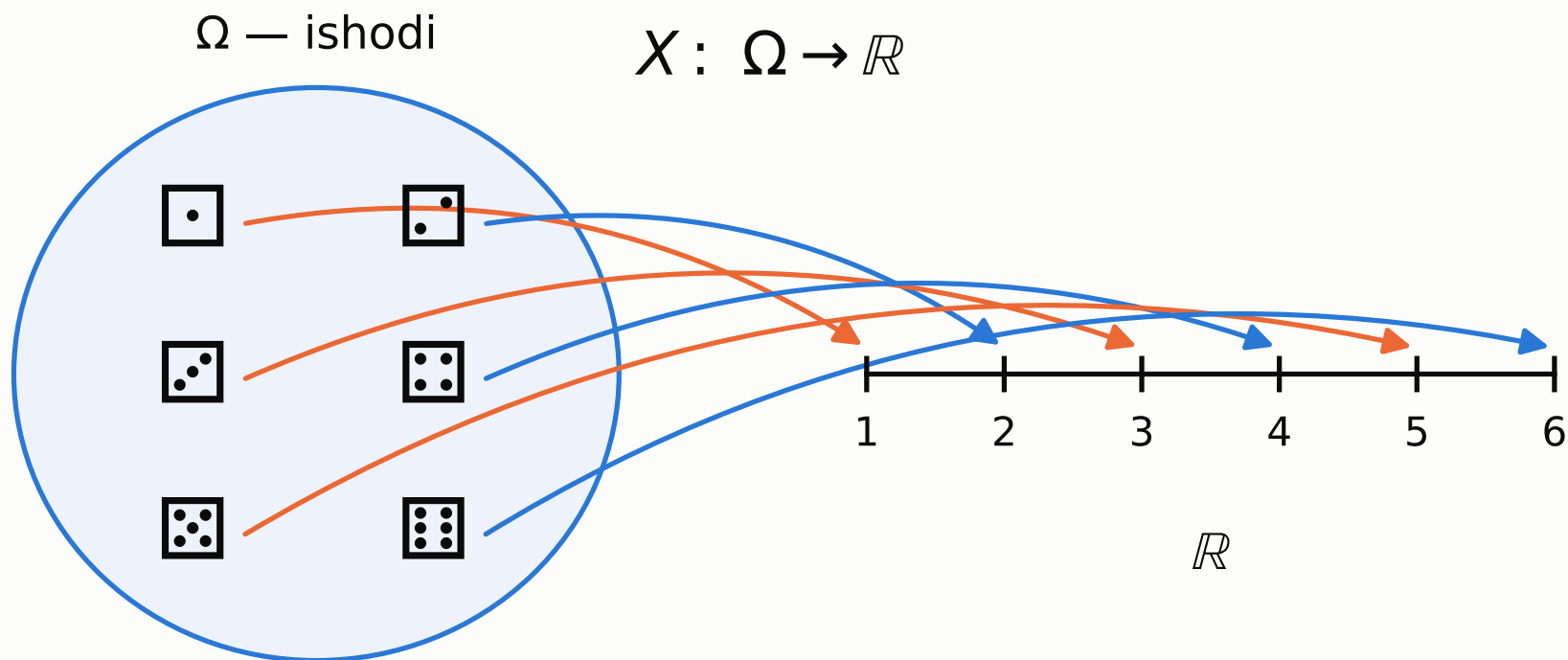
b — sistematska greška, ε — slučajna greška

Da bismo razumeli ε treba nam **jezik slučajnosti**.

II

Jezik slučajnosti

Slučajna promenljiva



$$P(X = k) \geq 0, \quad \sum_k P(X = k) = 1$$

Neprekidna promenljiva

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$$

f — gustina, F — funkcija raspodele

Očekivanje i varijansa

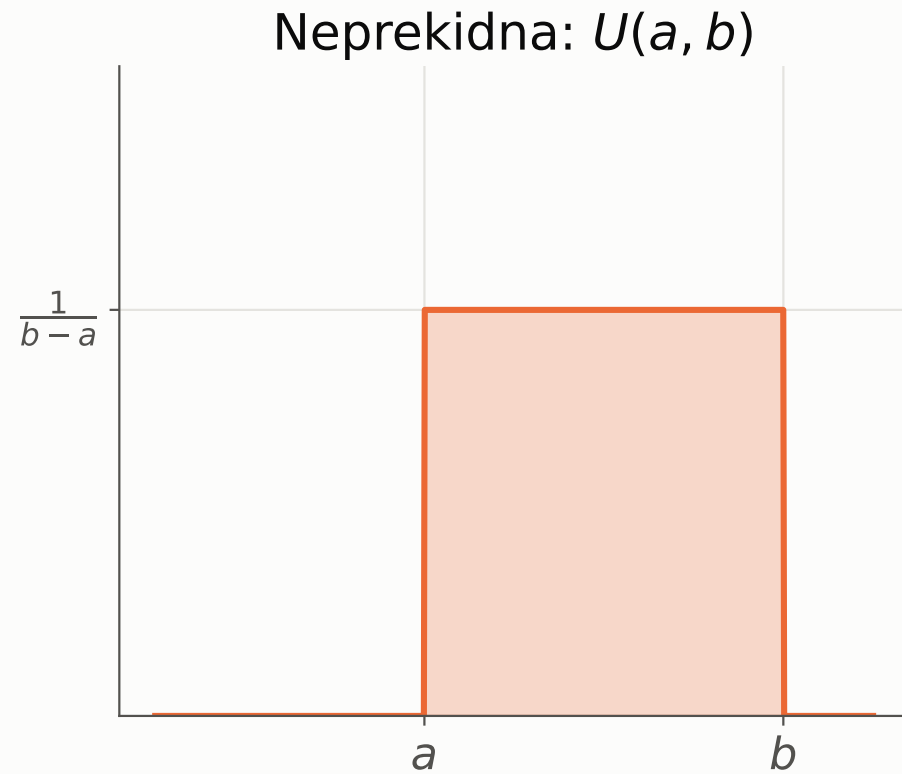
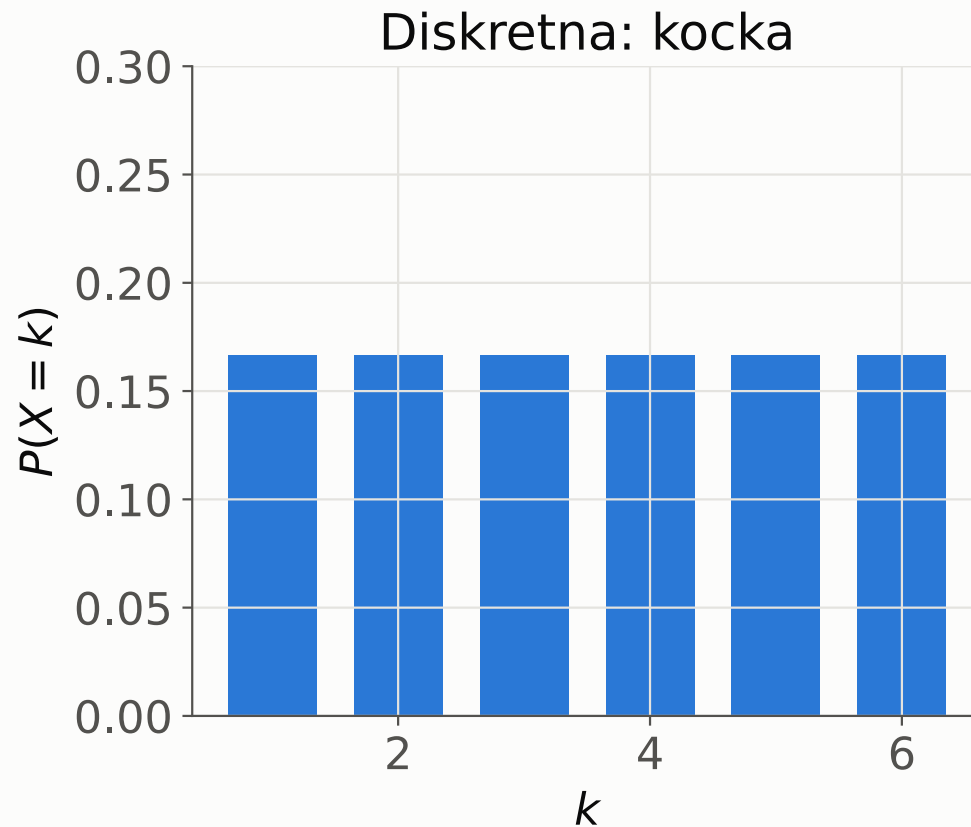
$$E[X] = \sum_k k P(X = k) \qquad E[X] = \int x f(x) dx$$

$$\text{Var}(X) = E[(X - E[X])^2], \qquad \sigma = \sqrt{\text{Var}(X)}$$

Linearnost — važi uvek, čak i za zavisne:

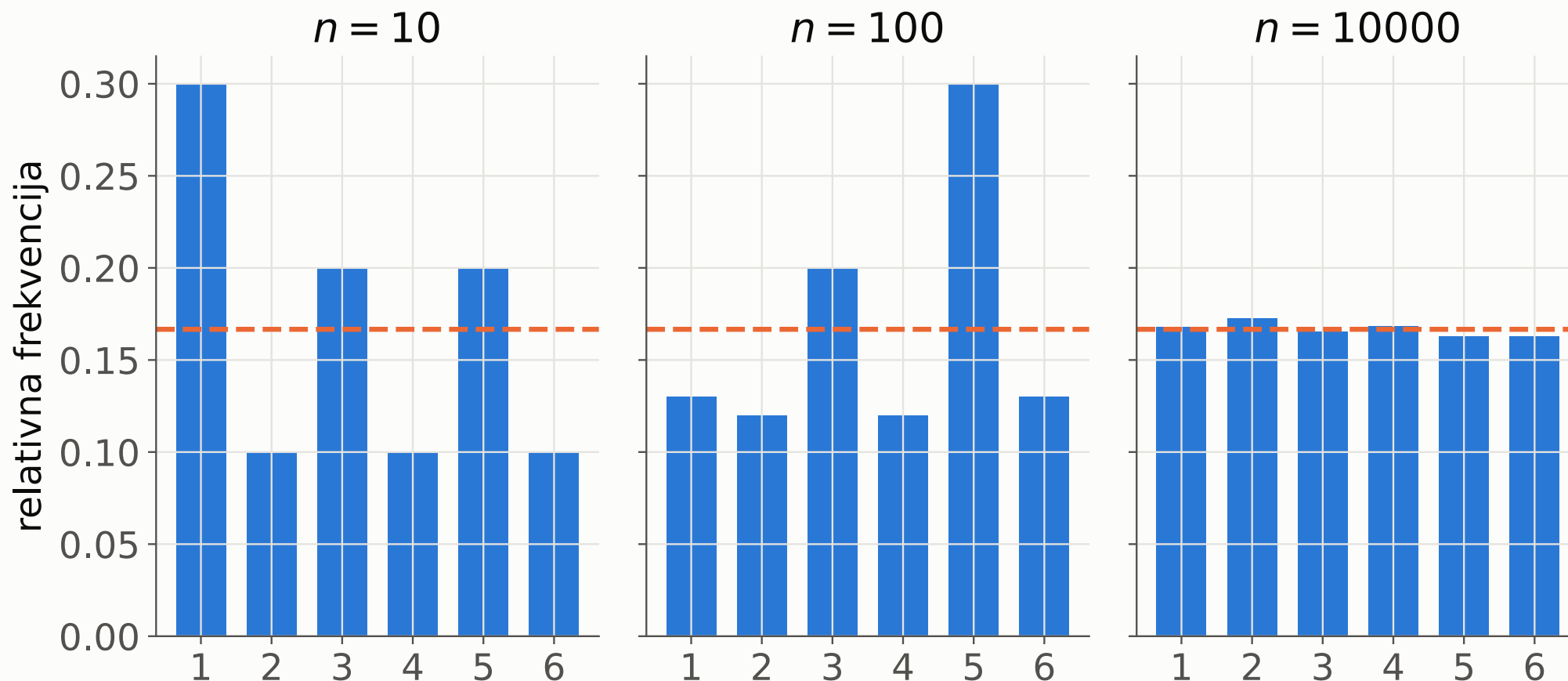
$$E[X + Y] = E[X] + E[Y]$$

Uniformna raspodela



$$X \sim U(a, b) : \quad E[X] = \frac{a+b}{2}, \quad \text{Var}(X) = \frac{(b-a)^2}{12}$$

Zakon velikih brojeva



$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{n \rightarrow \infty} E[X]$$

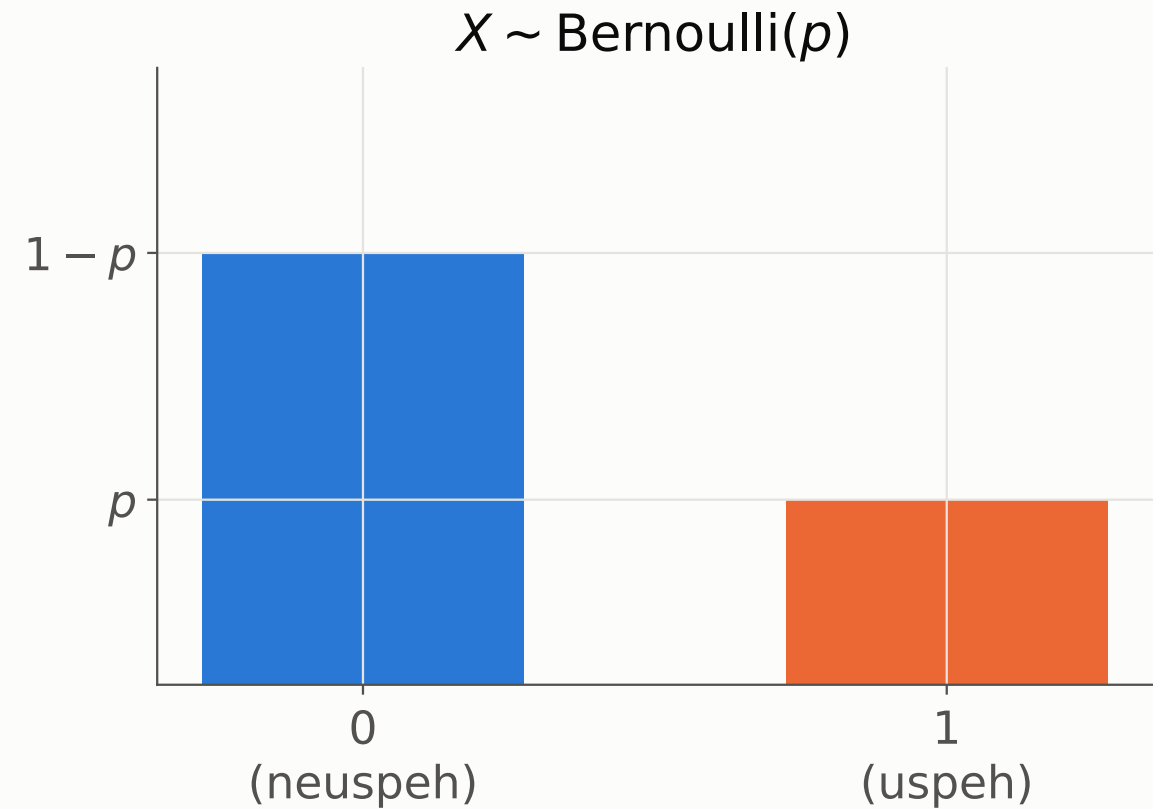
Binarna raspodela

Jedan pokušaj: uspeh ili neuspeh.

$$P(X = 1) = p$$

$$P(X = 0) = 1 - p$$

$$E[X] = p, \quad \text{Var}(X) = p(1 - p)$$



Indikator promenljiva

$$I_A = \begin{cases} 1, & \text{desio se } A \\ 0, & \text{u suprotnom} \end{cases}$$

$$\boxed{E[I_A] = P(A)}$$

Brojanje = zbir indikatora. Ovo ćemo koristiti za **quicksort**.

Binomna raspodela

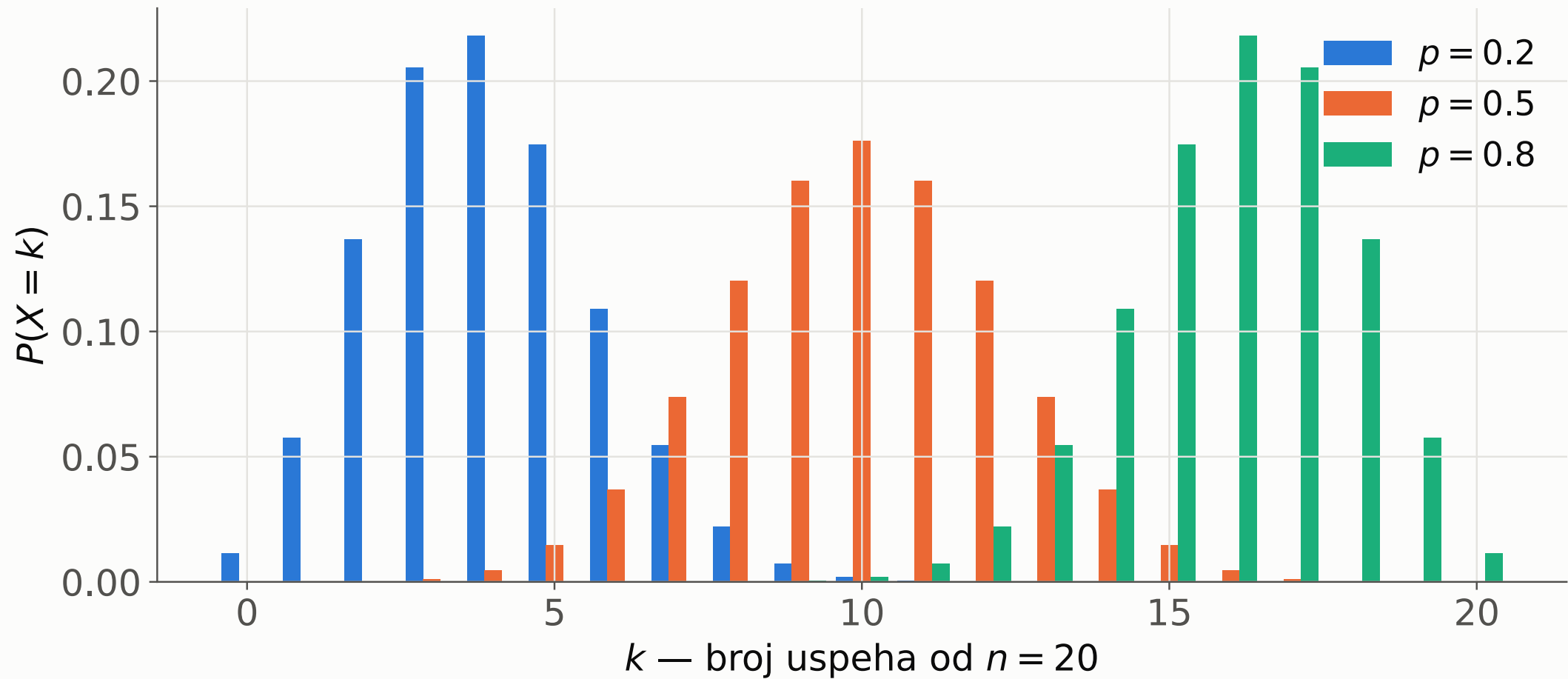
n nezavisnih binarnih pokušaja:

$$X = \sum_{i=1}^n X_i, \quad X_i \sim \text{Bernoulli}(p)$$

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

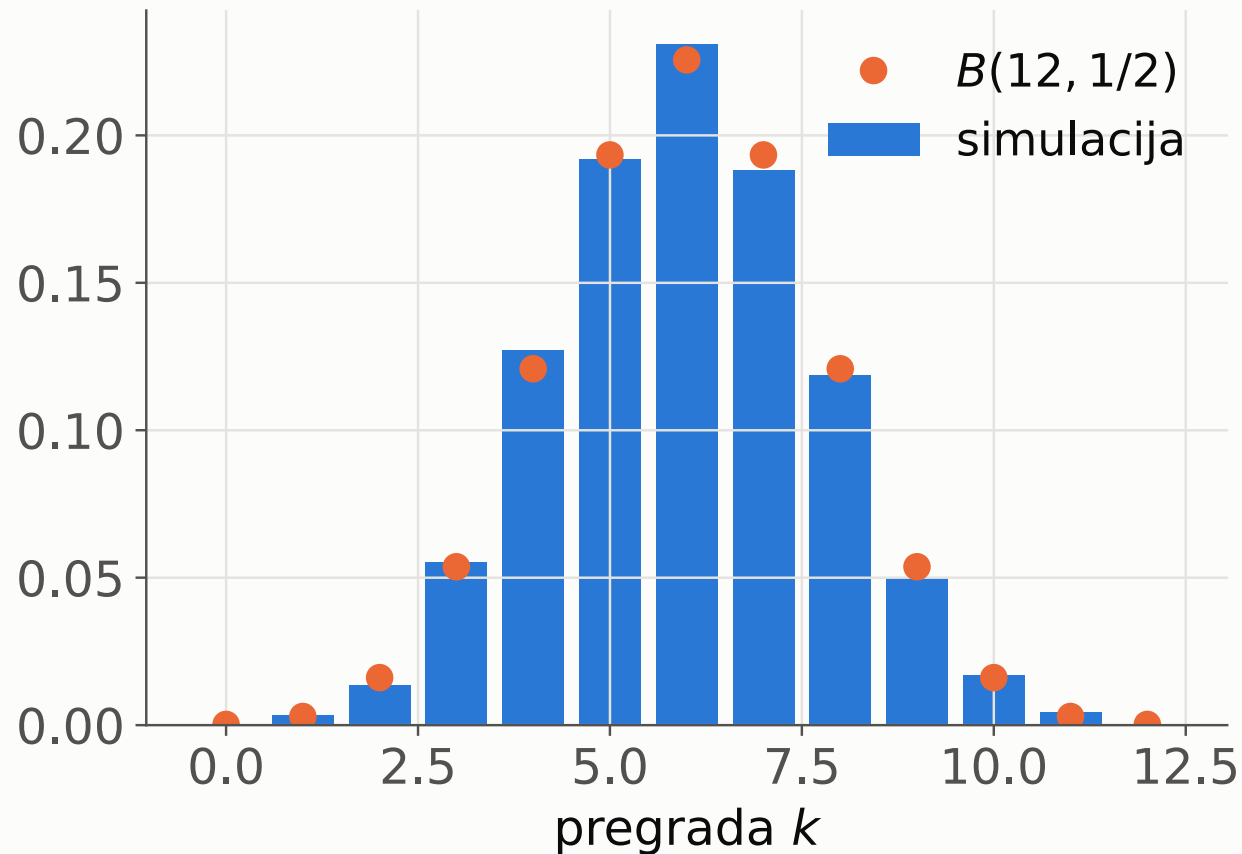
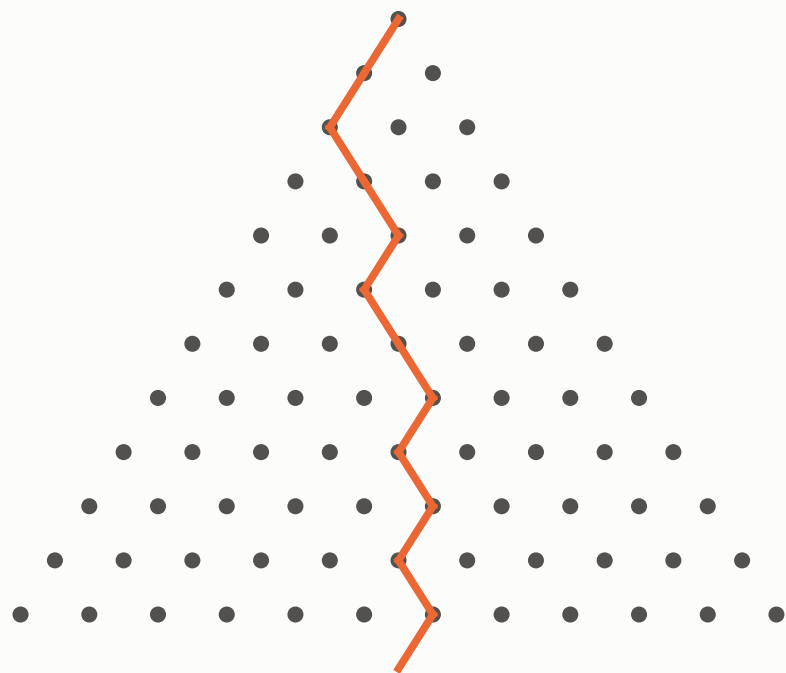
$$E[X] = np, \quad \text{Var}(X) = np(1 - p)$$

Binomna raspodela



Galtonova tabla

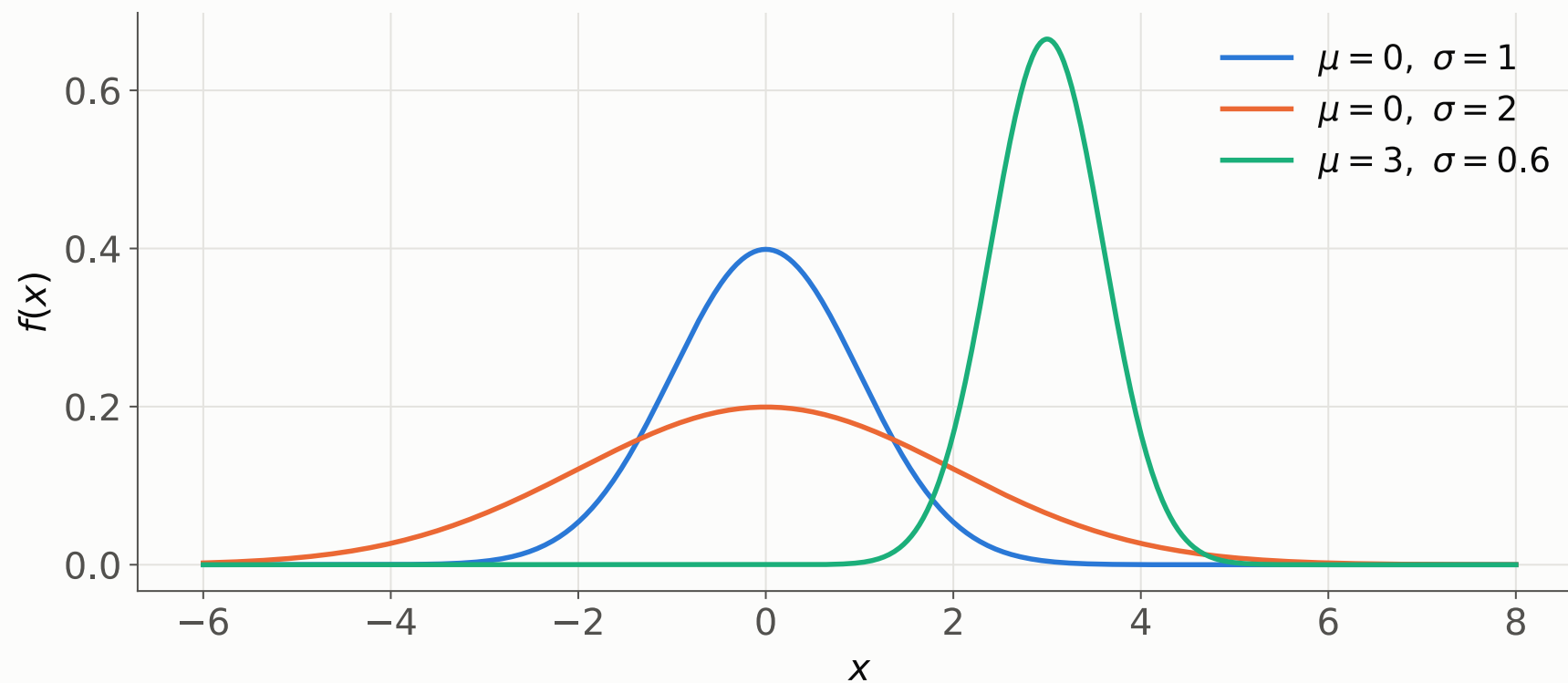
Galtonova tabla



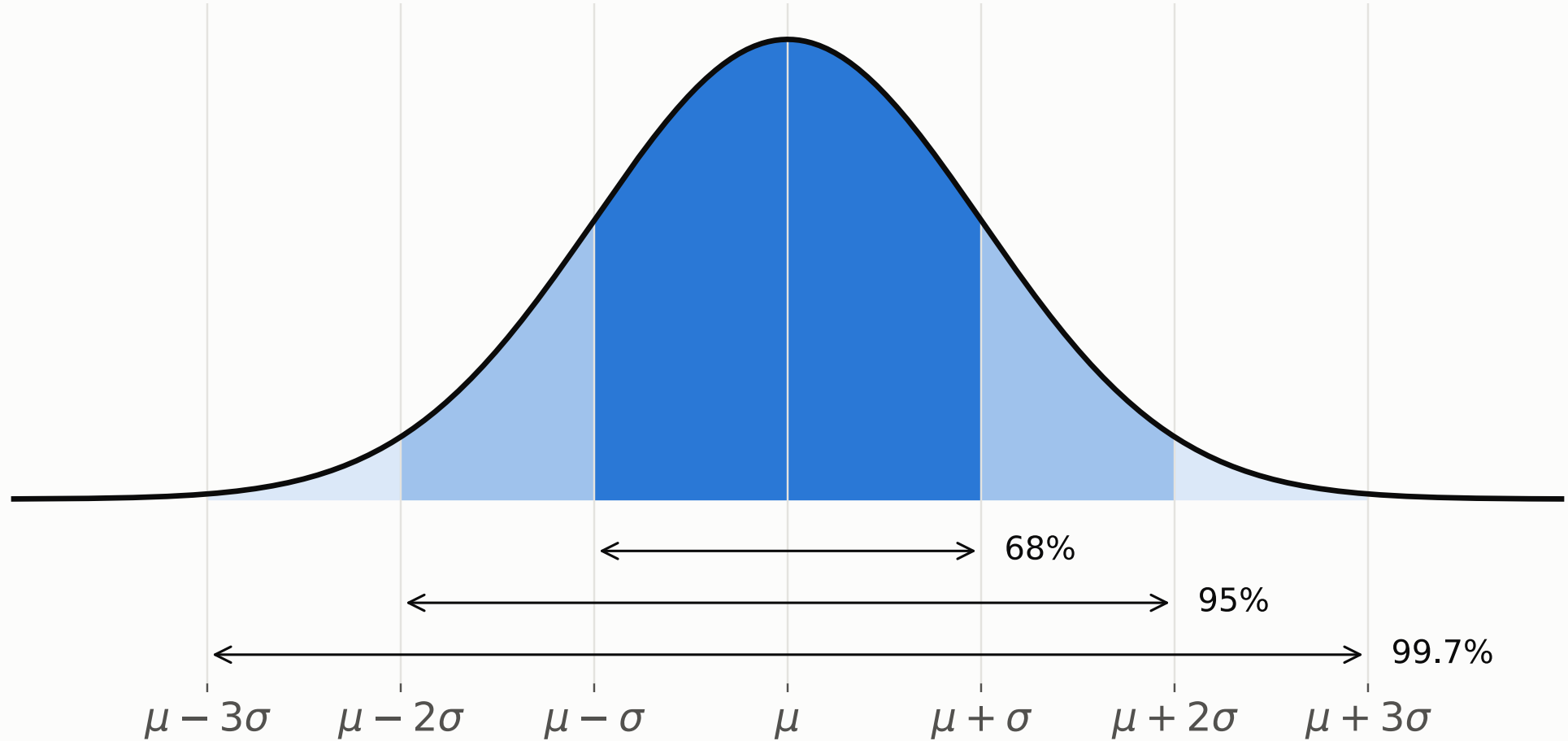
Levo ili desno, 12 puta: $k \sim B(12, \frac{1}{2})$

Normalna raspodela

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$



Pravilo 68 – 95 – 99.7



$$P(|X - \mu| \leq 2\sigma) \approx 0.95$$

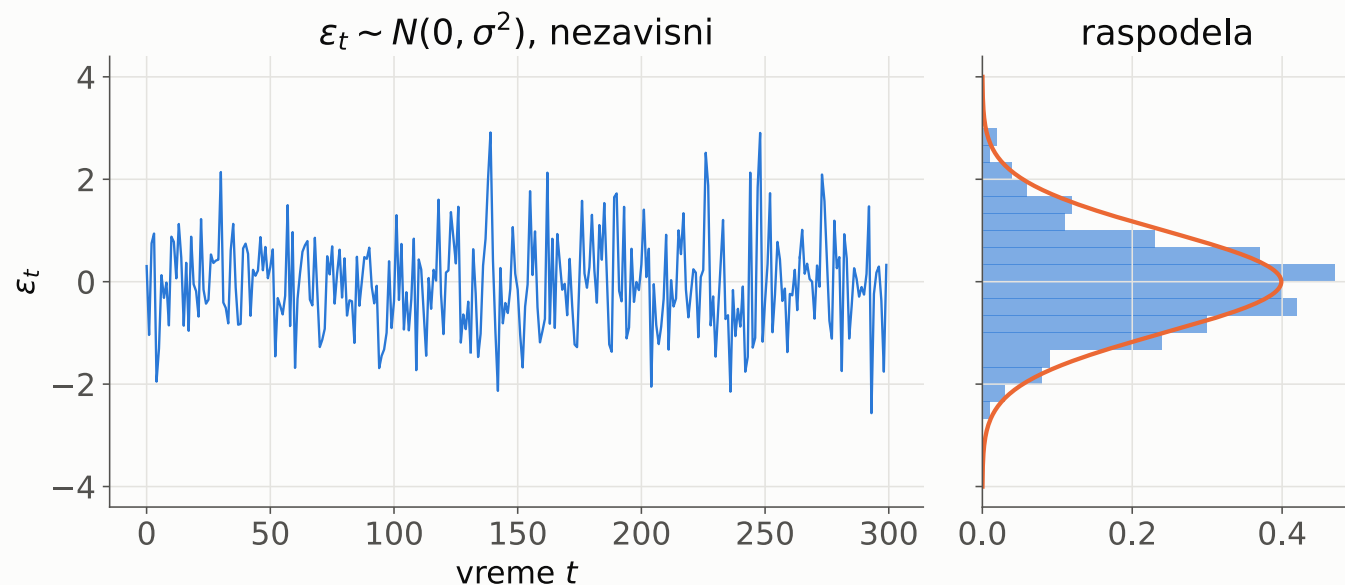
Beli šum

$$E[\varepsilon_t] = 0$$

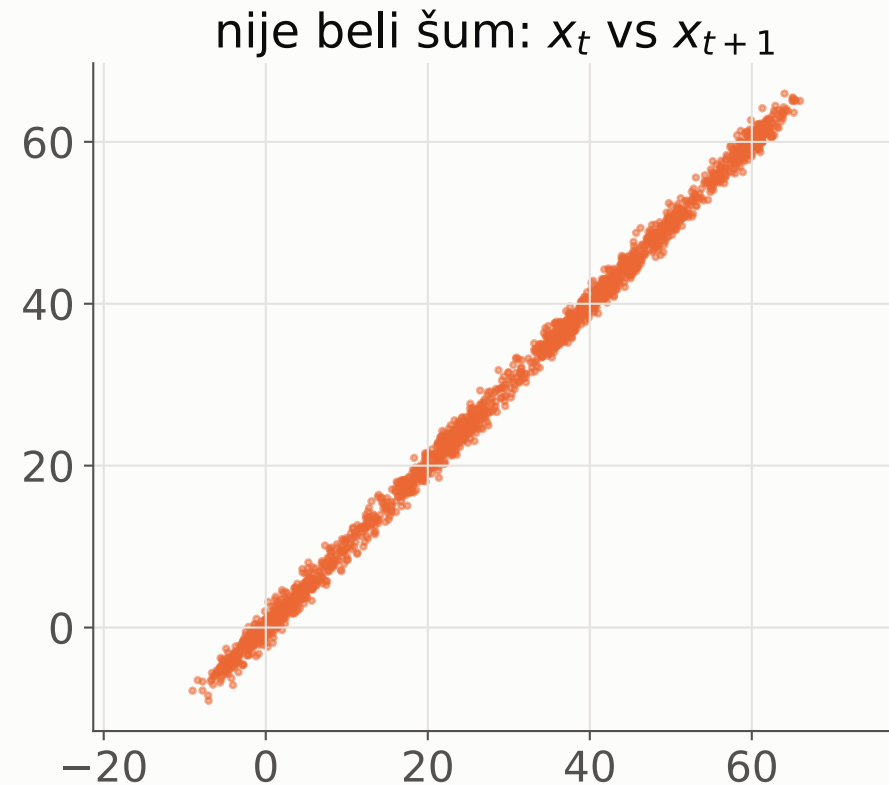
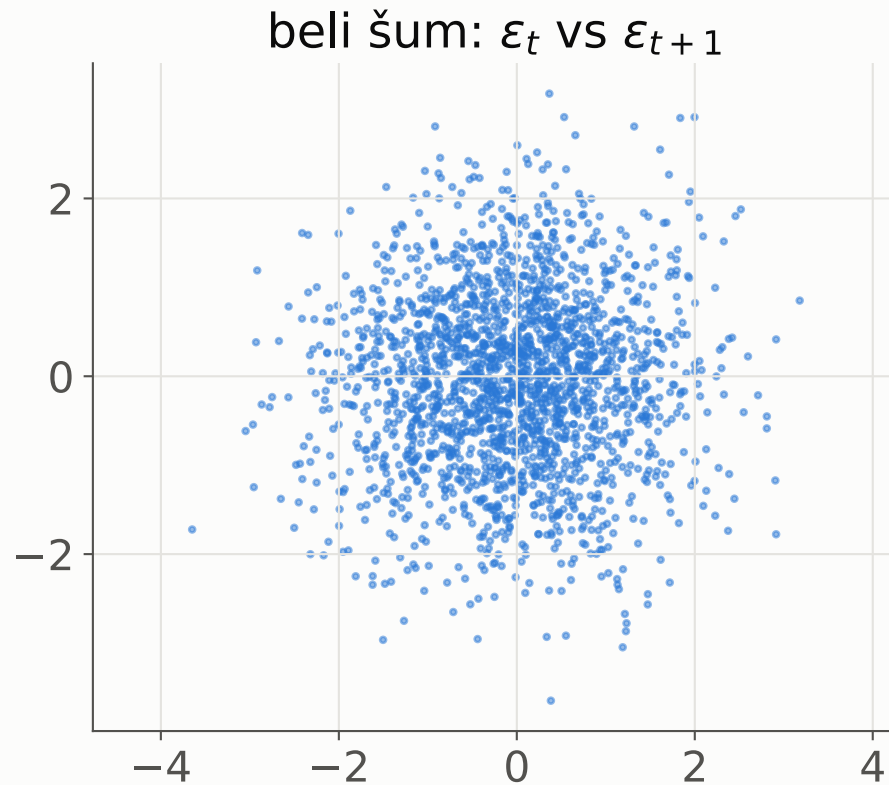
$$\text{Var}(\varepsilon_t) = \sigma^2$$

$$\text{Cov}(\varepsilon_t, \varepsilon_s) = 0, \quad t \neq s$$

Prošlost ne govori ništa o
budućnosti.



Beli šum ili ne?



$$x_{t+1} = x_t + \varepsilon_t$$

Slučajni hod **ima memoriju**.

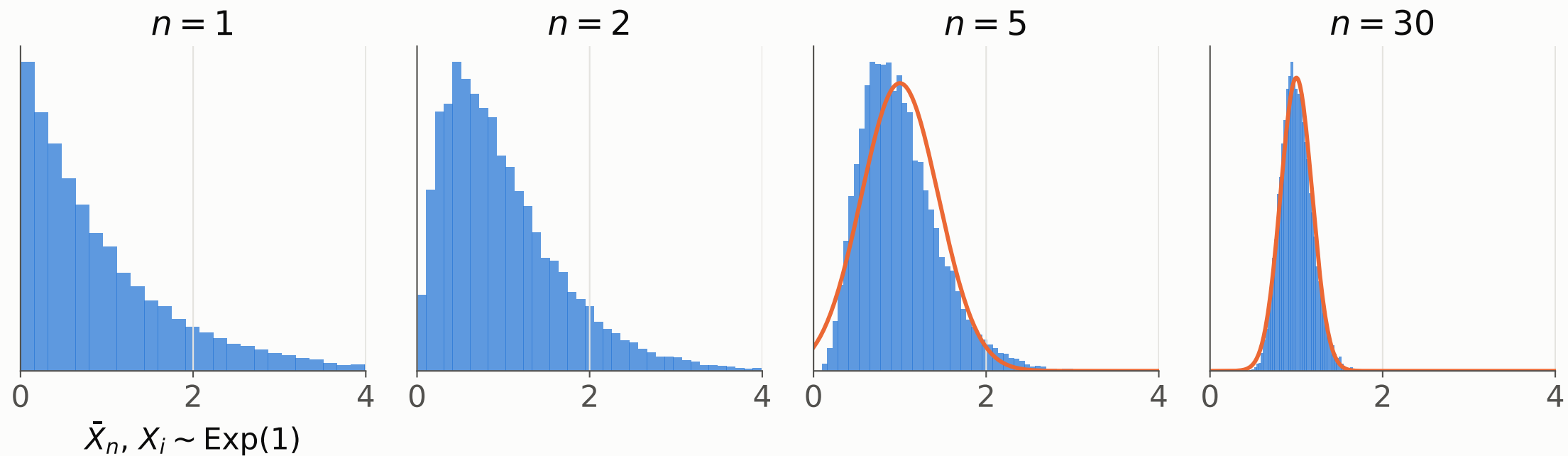
Centralna granična teorema

$X_1, \dots, X_n$ nezavisne, iste raspodele, $E[X_i] = \mu$, $\text{Var}(X_i) = \sigma^2$:

$$\frac{\bar{X}_n - \mu}{\sigma / \sqrt{n}} \xrightarrow{d} N(0, 1)$$

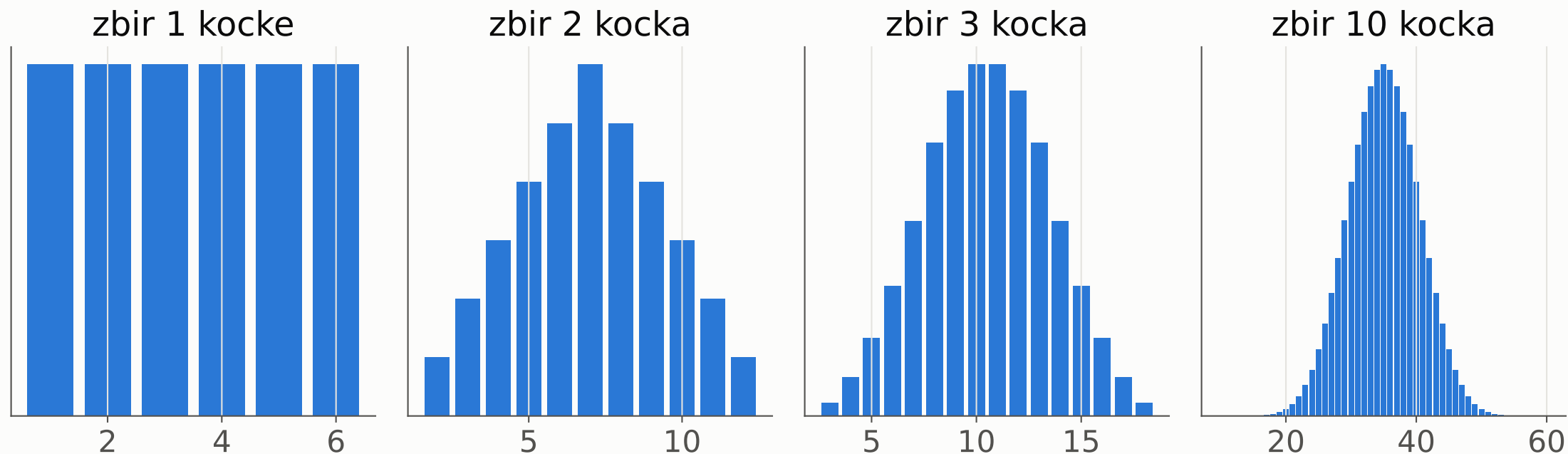
Bez obzira na **originalnu** raspodelu!

CGT: prosek eksponencijalnih



Krivo $\rightarrow$ simetrično $\rightarrow$ normalno

CGT: zbir kockica



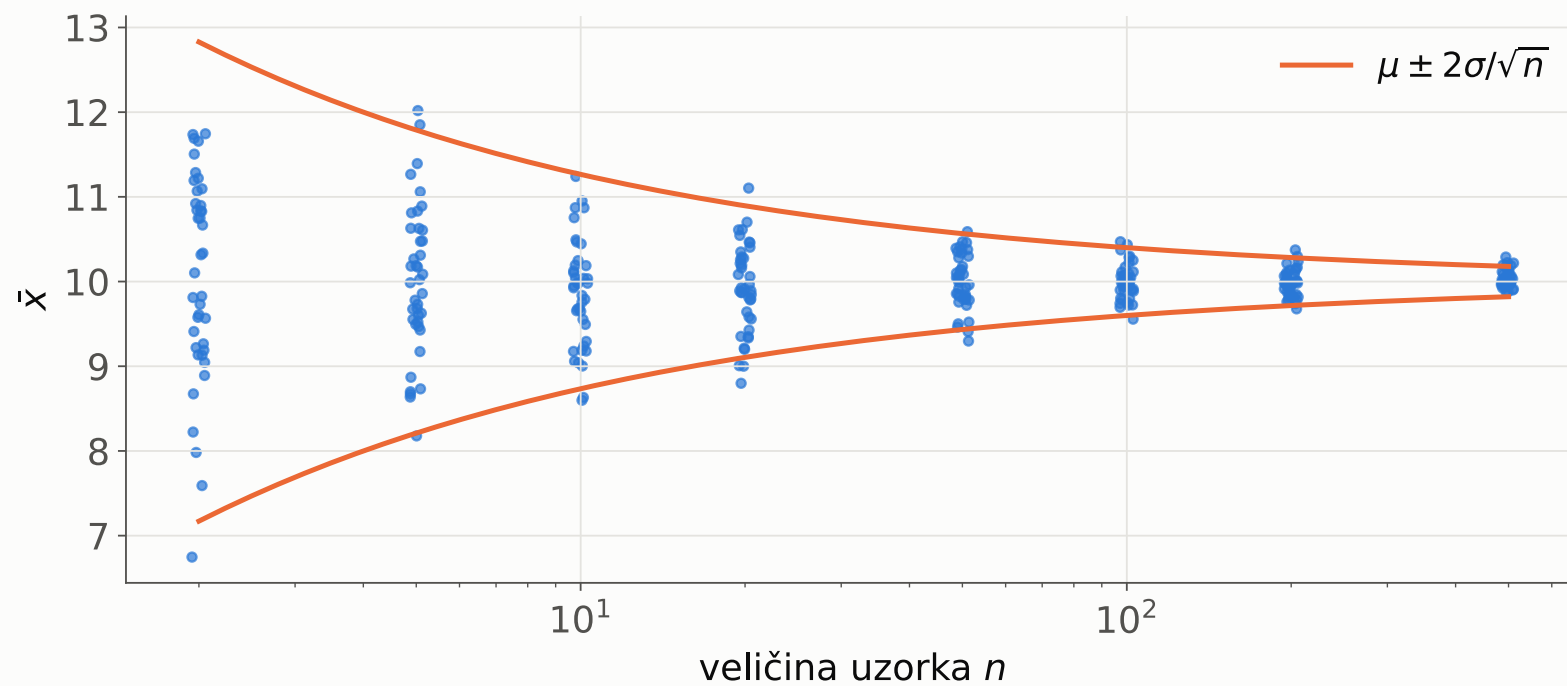
$$\varepsilon = \delta_1 + \delta_2 + \cdots + \delta_m \Rightarrow \varepsilon \approx N(0, \sigma^2)$$

Šum je zbir mnogo malih uzroka — zato je **normalan**.

III

Merenja i testovi

Standardna greška



$$SE(\bar{x}) = \frac{\sigma}{\sqrt{n}}$$

4× više merenja $\Rightarrow$ 2× preciznije

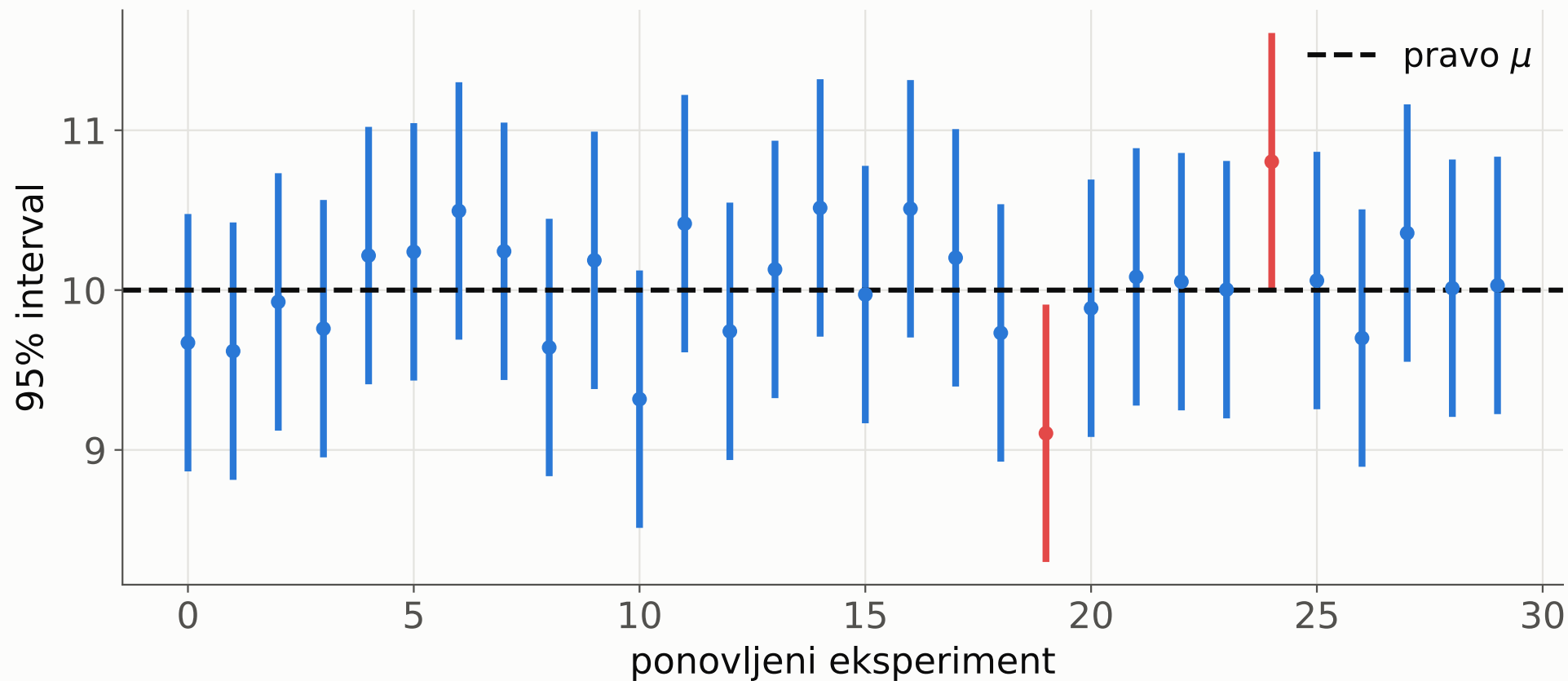
Koliko ponavljanja?

Želimo grešku najviše Δ sa sigurnošću 95%:

$$1.96 \frac{\sigma}{\sqrt{n}} \leq \Delta \implies n \geq \left(\frac{1.96 \sigma}{\Delta} \right)^2$$

$$\sigma = 2, \Delta = 0.5 \implies n \geq 62$$

Interval poverenja



$$\bar{x} \pm 1.96 \frac{s}{\sqrt{n}}$$

pogodi μ u $\approx 95\%$ eksperimenata

Test hipoteze

1. postavi H_0 i H_1
2. izaberi prag α (npr. 0.05)
3. izračunaj statistiku iz podataka
4. izračunaj p -vrednost
5. $p < \alpha \Rightarrow$ odbaci H_0

t-test

Jedan uzorak:

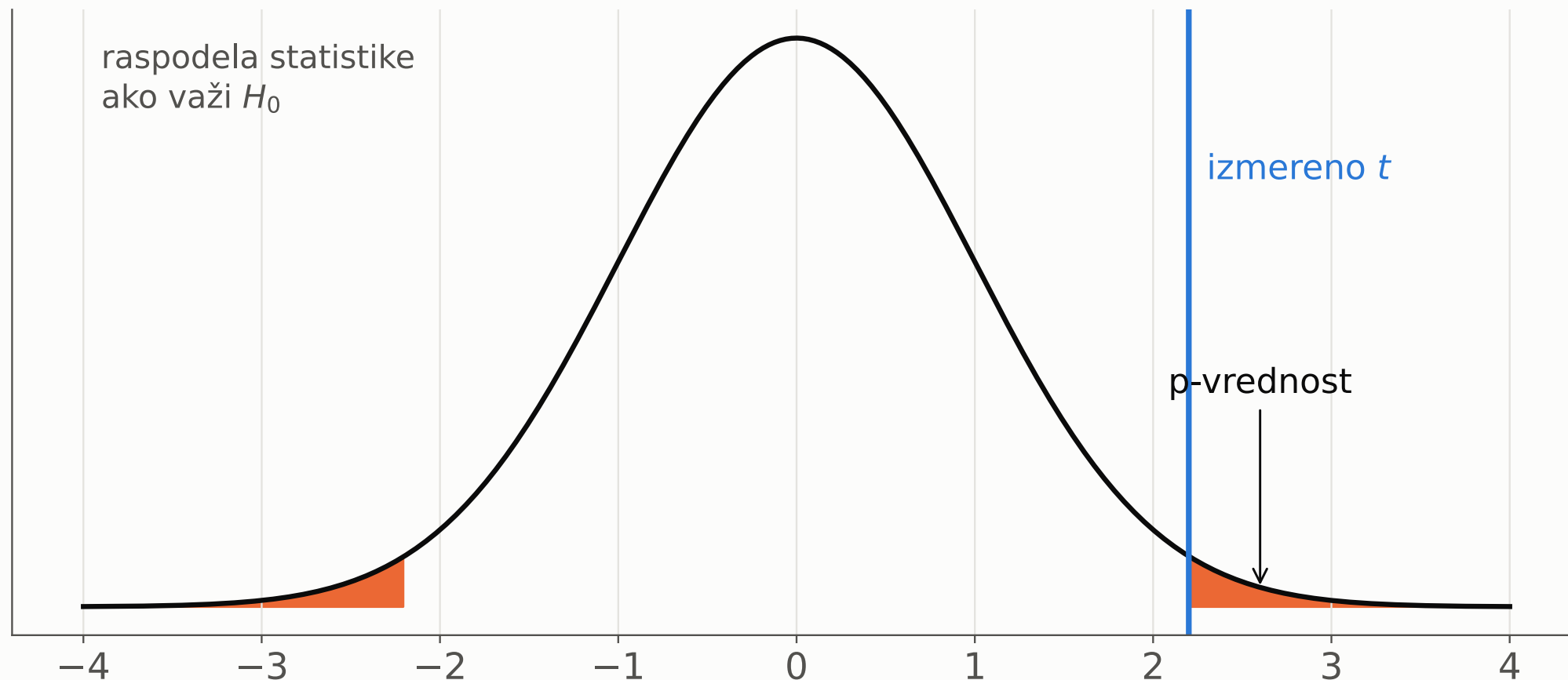
$$t = \frac{\bar{x} - \mu_0}{s / \sqrt{n}}$$

Dva uzorka (Welch):

$$t = \frac{\bar{x}_A - \bar{x}_B}{\sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}}}$$

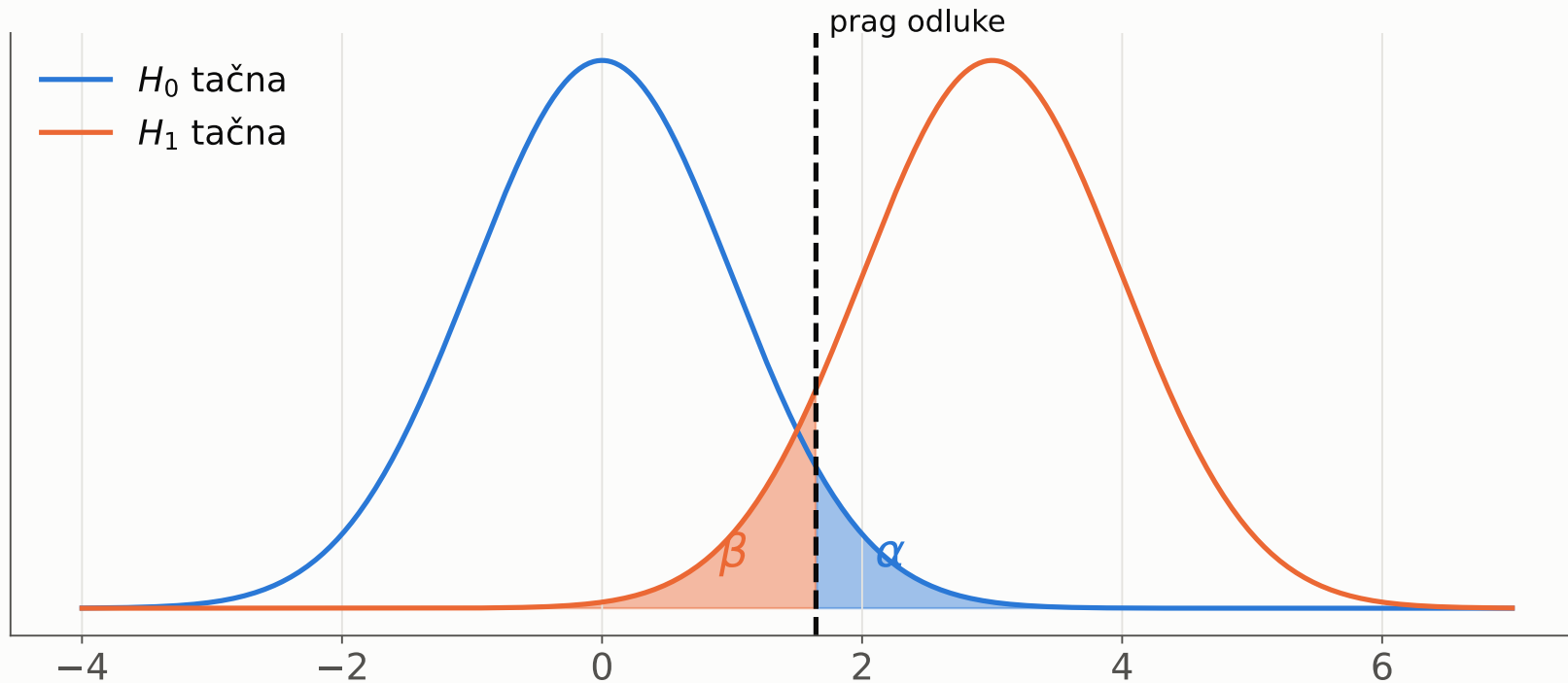
signal / šum

p-vrednost



$$p = P(|T| \geq |t| \mid H_0)$$

Greške I i II vrste



	H_0 tačna	H_0 netačna
odbacimo H_0	greška I vrste (α)	✓ moć $1 - \beta$
ne odbacimo	✓	greška II vrste (β)

Zamka: mnogo testova

Test m nezavisnih hipoteza, sve H_0 tačne:

Verovatnoća bar jednog lažnog otkrića:

$$1 - (1 - \alpha)^m$$

m	1	5	20	100
verovatnoća	5%	23%	64%	99%

Bonferoni: koristi α/m

IV

Slučajni algoritmi

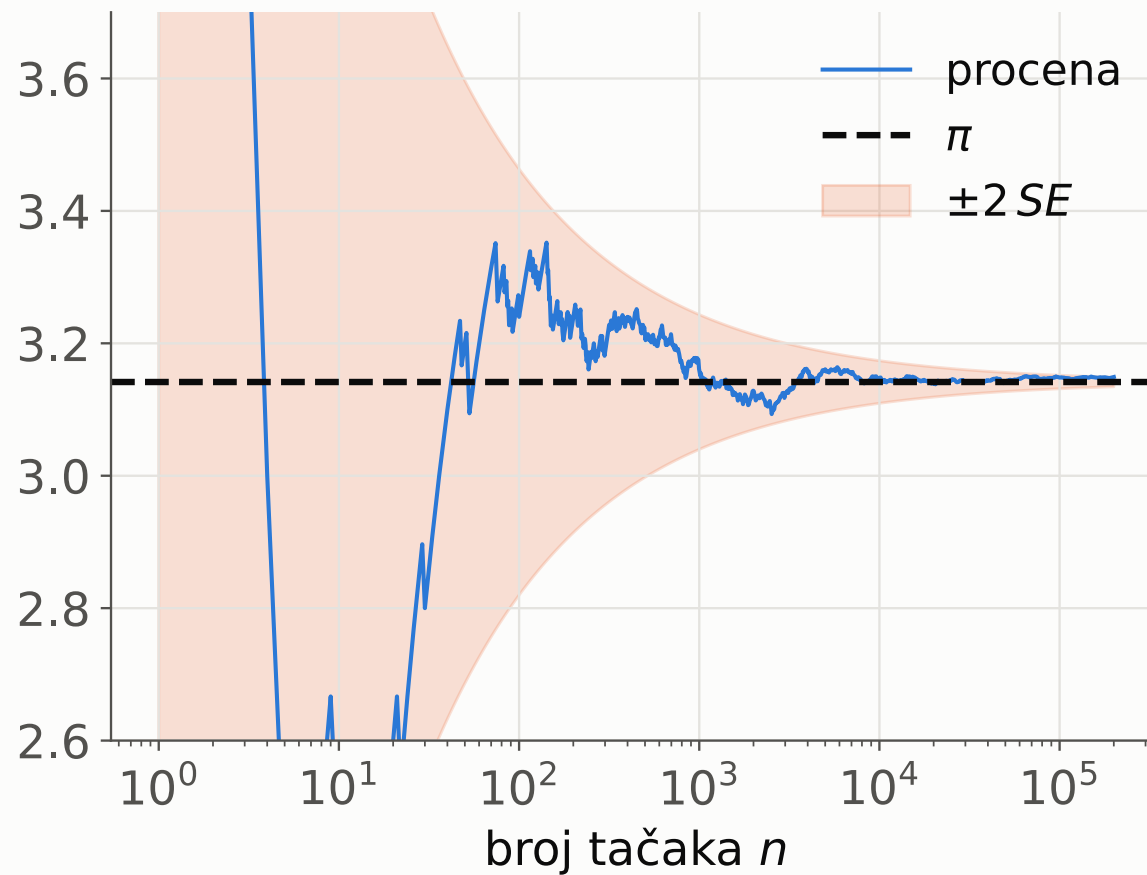
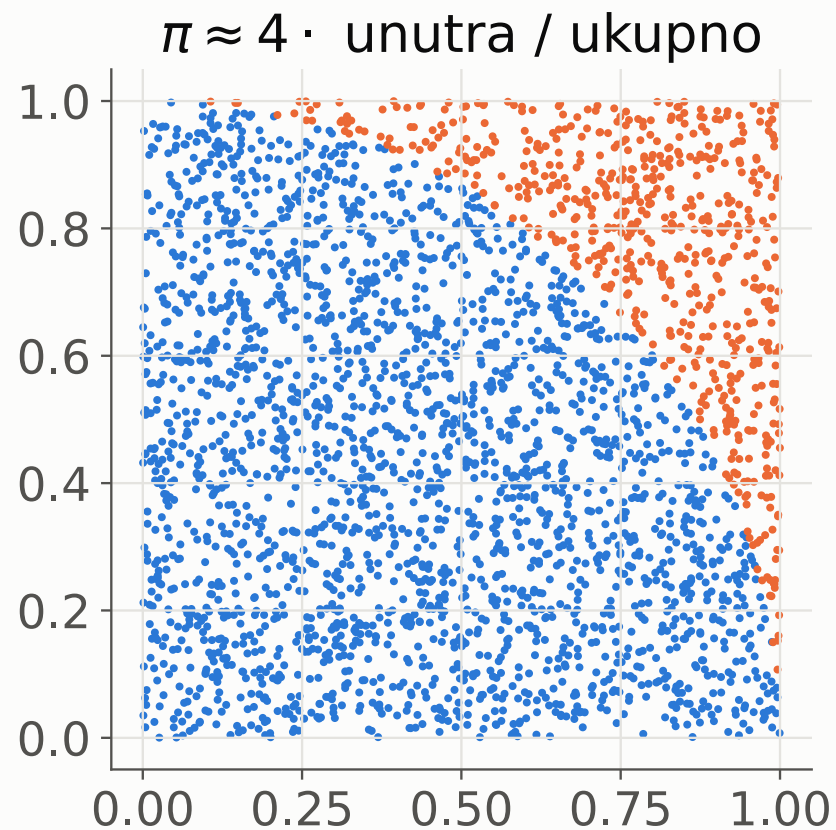
Algoritam koji baca kockicu

$$A(x) \longrightarrow A(x, r), \quad r \sim U$$

	Las Vegas	Monte Carlo
rezultat	uvek tačan	tačan sa verovatnoćom
vreme	slučajno	ograničeno
primer	quicksort	procena π

Analiza: **indikatori** + **linearnost očekivanja**

Monte Carlo: procena π

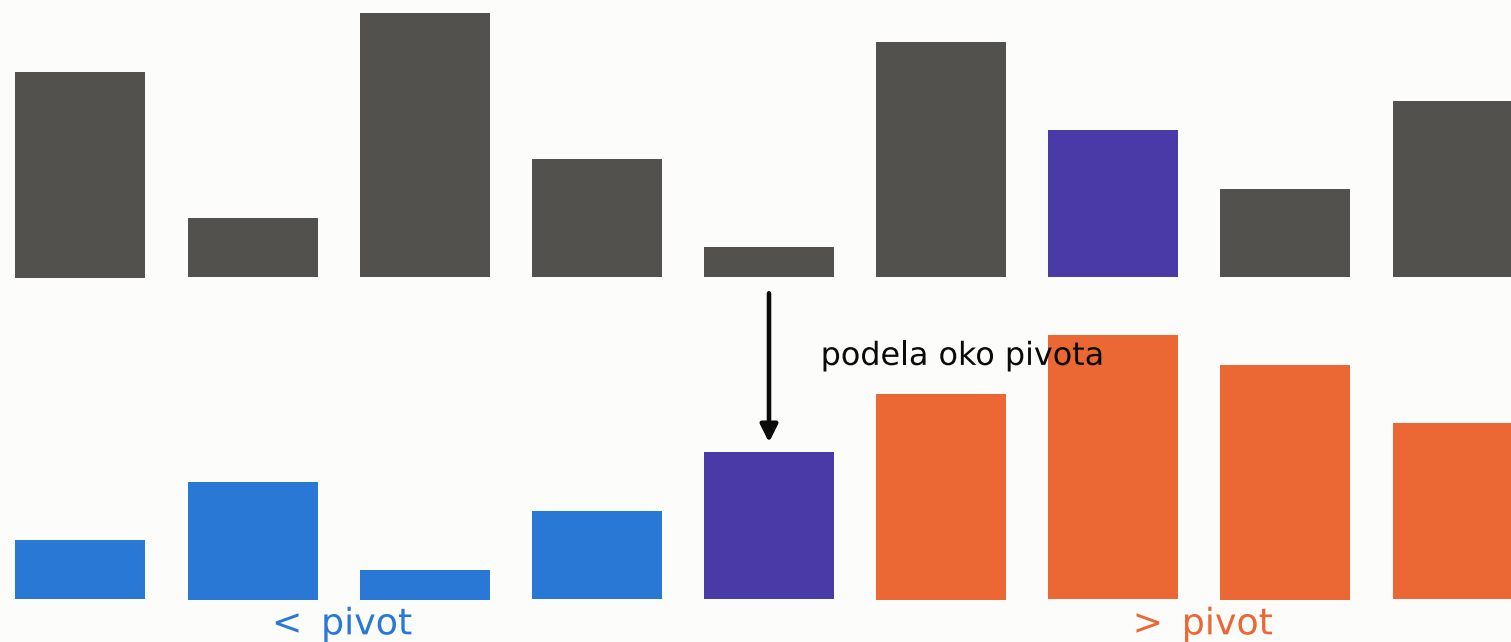


$$\hat{\pi} = \frac{4}{n} \sum_{i=1}^n I(x_i^2 + y_i^2 \leq 1), \quad SE = 4 \sqrt{\frac{p(1-p)}{n}}, \quad p = \frac{\pi}{4}$$

V

Istraživanje: Quicksort

Algoritam



```
quicksort(S):  
  y ← slučajan element iz S  
  S1 ← {s < y}, S2 ← {s > y}      # |S| - 1 poređenja  
  return quicksort(S1) + [y] + quicksort(S2)
```

Pitanje i hipoteze

$$\underbrace{\sim n \log_2 n}_{\text{najbolji}} \leq C_n \leq \underbrace{\binom{n}{2}}_{\text{najgori}}$$

H_1 : slučajan pivot $\Rightarrow E[C_n] \approx c \cdot n \ln n$

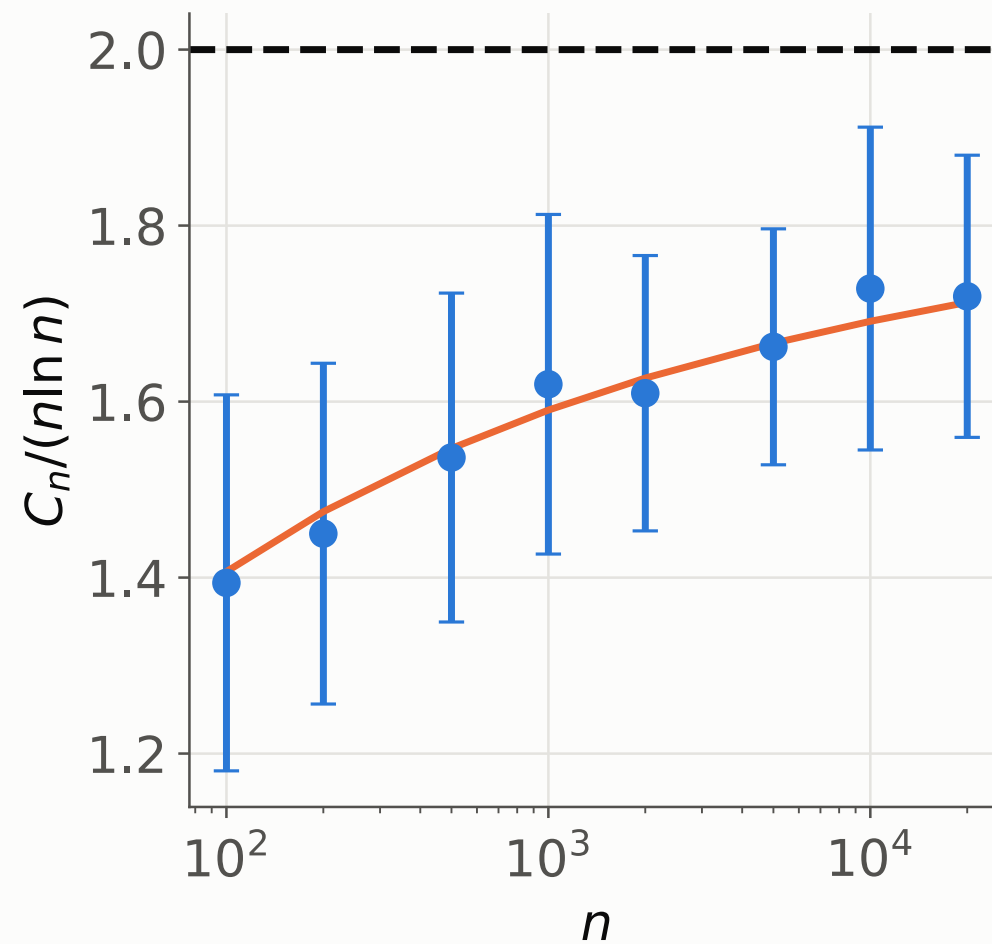
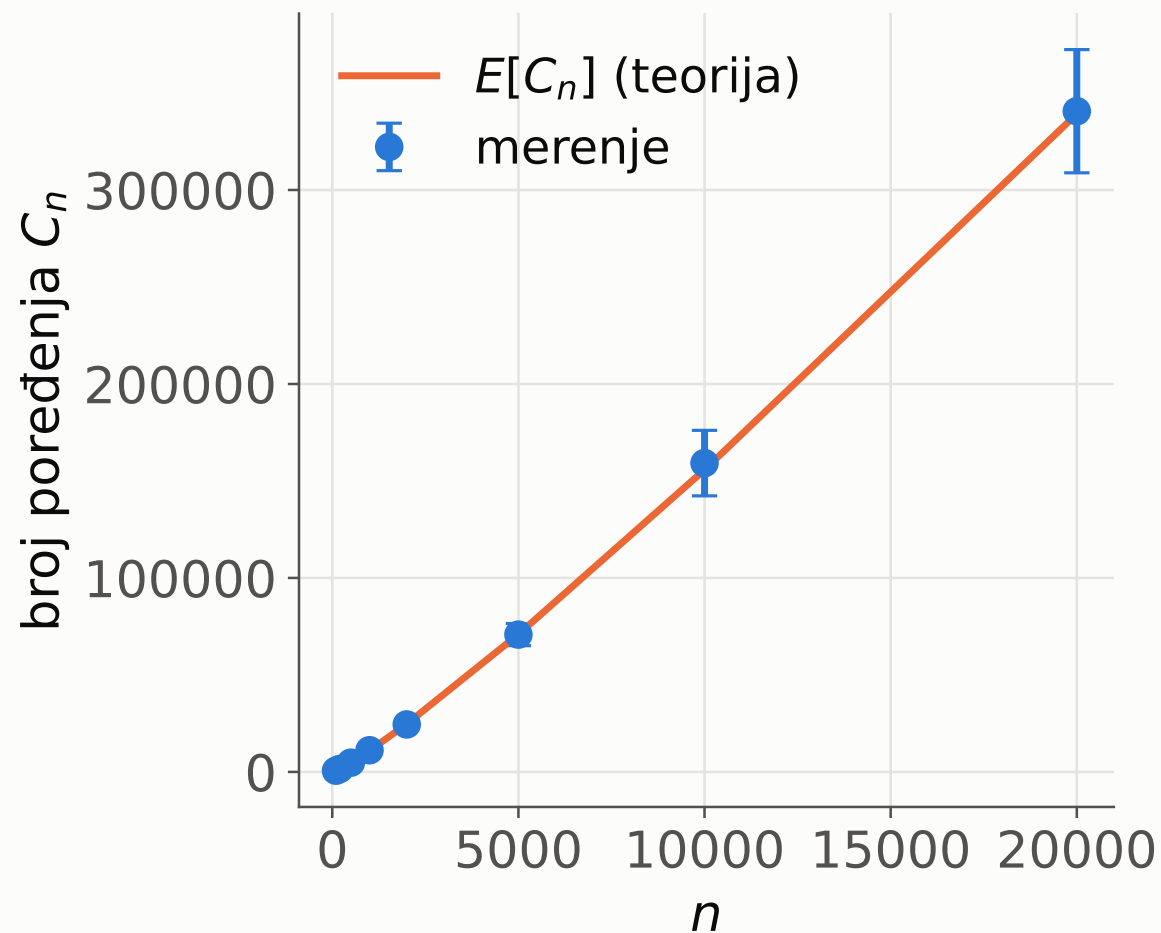
H_2 : prvi element kao pivot, sortirani ulaz $\Rightarrow C_n = \Theta(n^2)$

Dizajn eksperimenta

nezavisne	n , izbor pivota, tip ulaza
zavisna	broj poređenja C_n
kontrolne	isti kod, isti generator, fiksiran seed
ponavljanja	30 po svakom n
n	$10^2 \dots 2 \cdot 10^4$

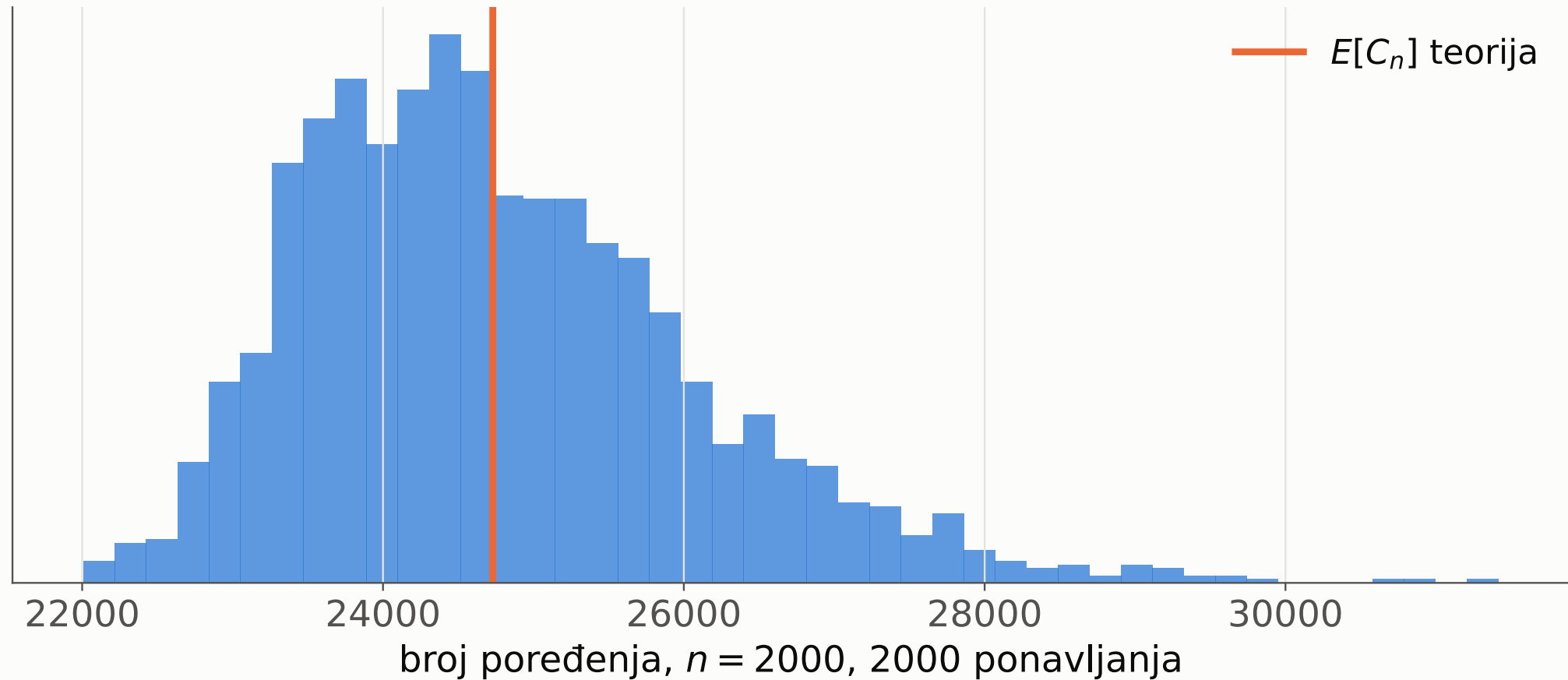
Zašto ne vreme? $T = c \cdot C_n + \varepsilon$ — keš, OS, CPU

Rezultati



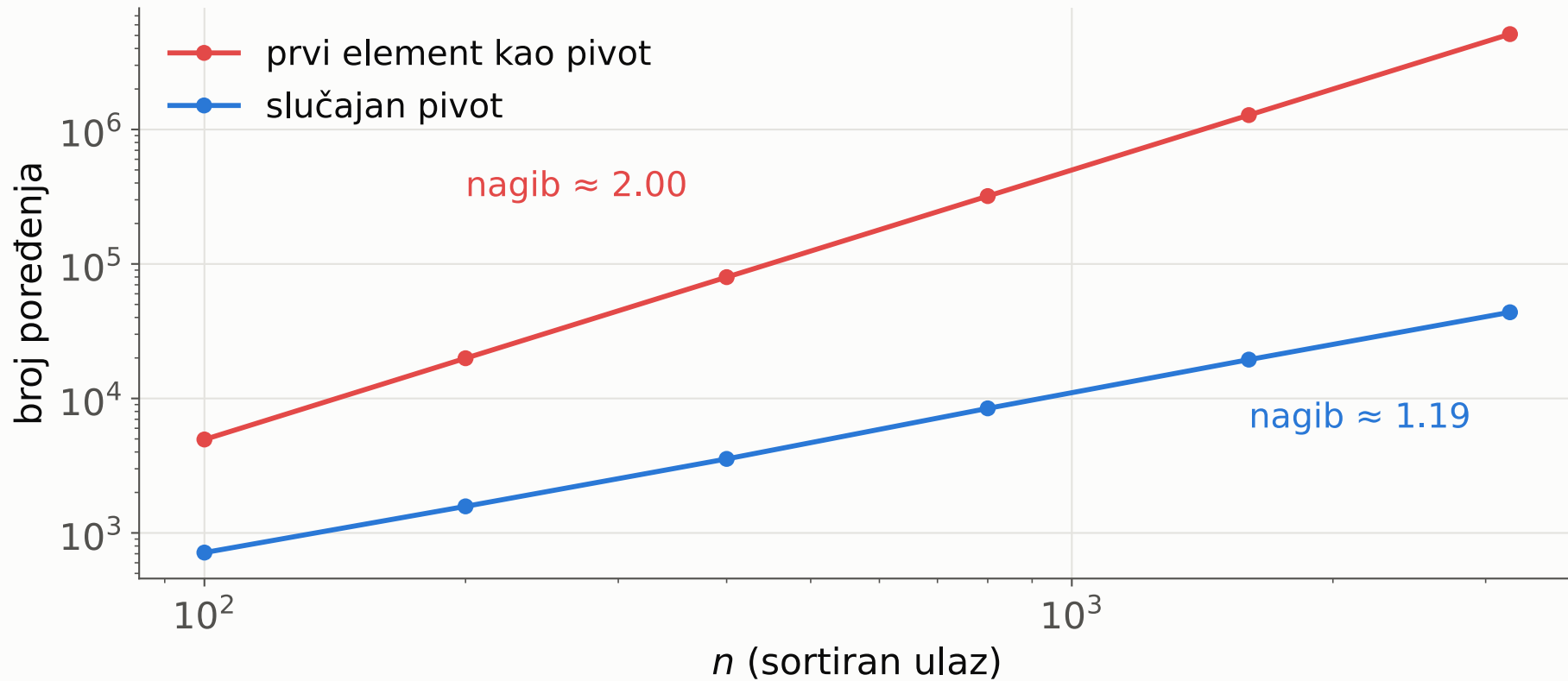
Merenja prate teoriju, ali $C_n / (n \ln n)$ sporo raste ka 2.

Raspodela C_n



Nije simetrična — C_n **nije** zbir nezavisnih promenljivih.

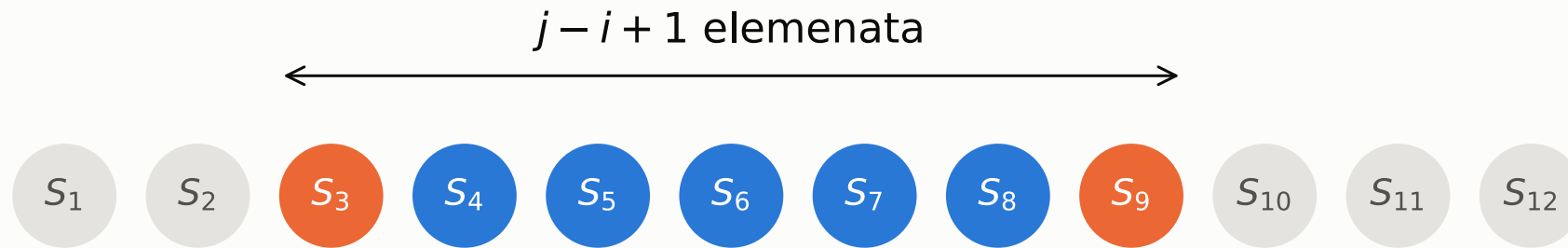
Log-log grafik



$$C = a n^k \implies \log C = \log a + k \log n$$

H_2 potvrđena: nagib 2.

Teorija: indikatori



poređeni $\Leftrightarrow$ prvi izabran pivot među njima je S_i ili S_j

$$X_{ij} = I(S_i \leftrightarrow S_j), \quad P(X_{ij} = 1) = \frac{2}{j - i + 1}$$

Teorija: izvođenje

$$E[C_n] = E\left[\sum_{i < j} X_{ij}\right] = \sum_{i < j} E[X_{ij}] = \sum_{i < j} \frac{2}{j - i + 1}$$

$$= 2(n + 1)H_n - 4n, \quad H_n = \sum_{k=1}^n \frac{1}{k} \approx \ln n$$

$$E[C_n] \approx 2n \ln n - 2.85 n$$

Zaključak eksperimenta

	Eksperiment	Teorija
rast	nagib $\approx 1.1\text{--}1.2$	$n \ln n$
konstanta	$C_n / (n \ln n) \rightarrow 1.7$	$\rightarrow 2$, ali sporo
sortiran ulaz	nagib 2.00	$\binom{n}{2}$

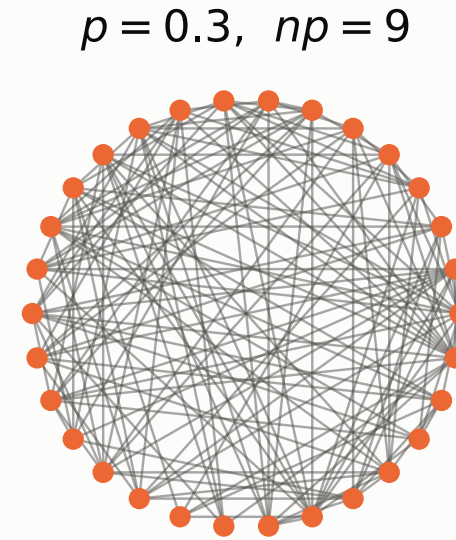
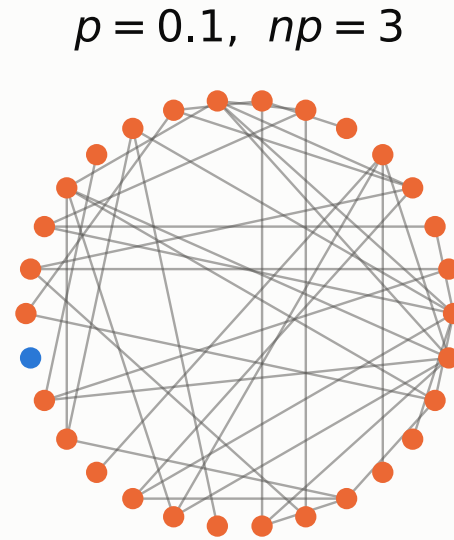
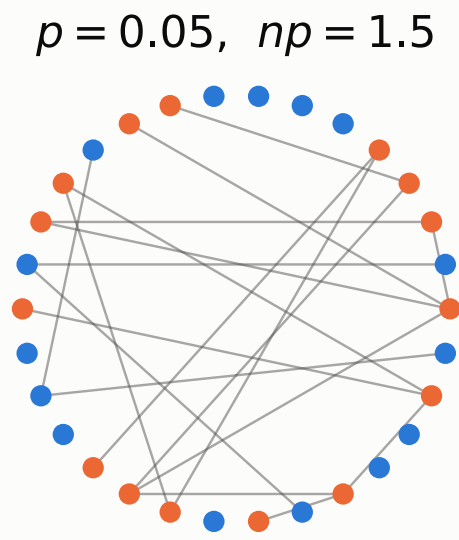
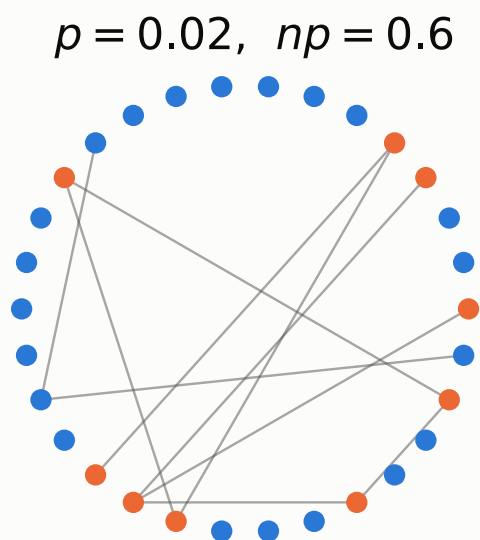
Razlika 1.7 vs 2? Član $-2.85 n$ — teorija **objašnjava** i odstupanje.

VI

Istraživanje: slučajni grafovi

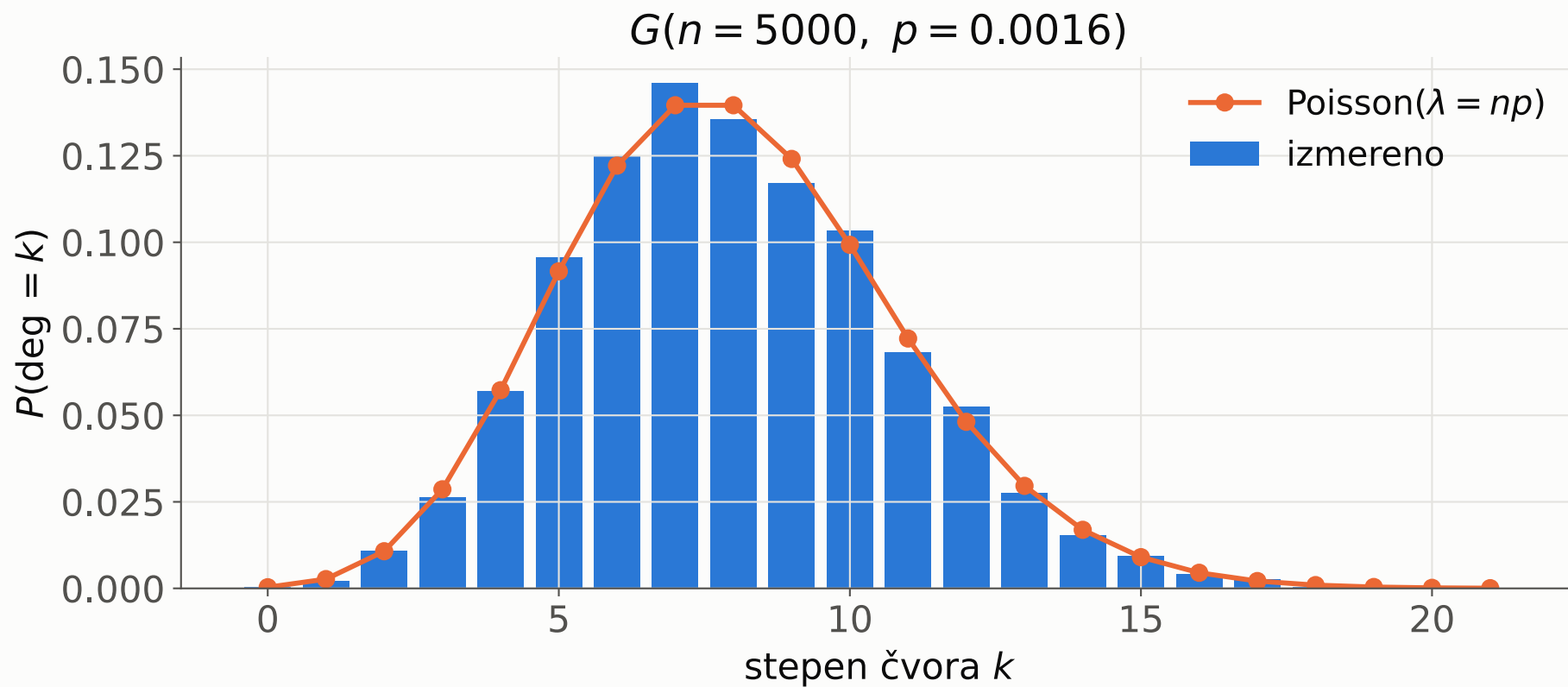
Erdős–Rényi graf $G(n, p)$

Svaka ivica postoji nezavisno sa verovatnoćom p .



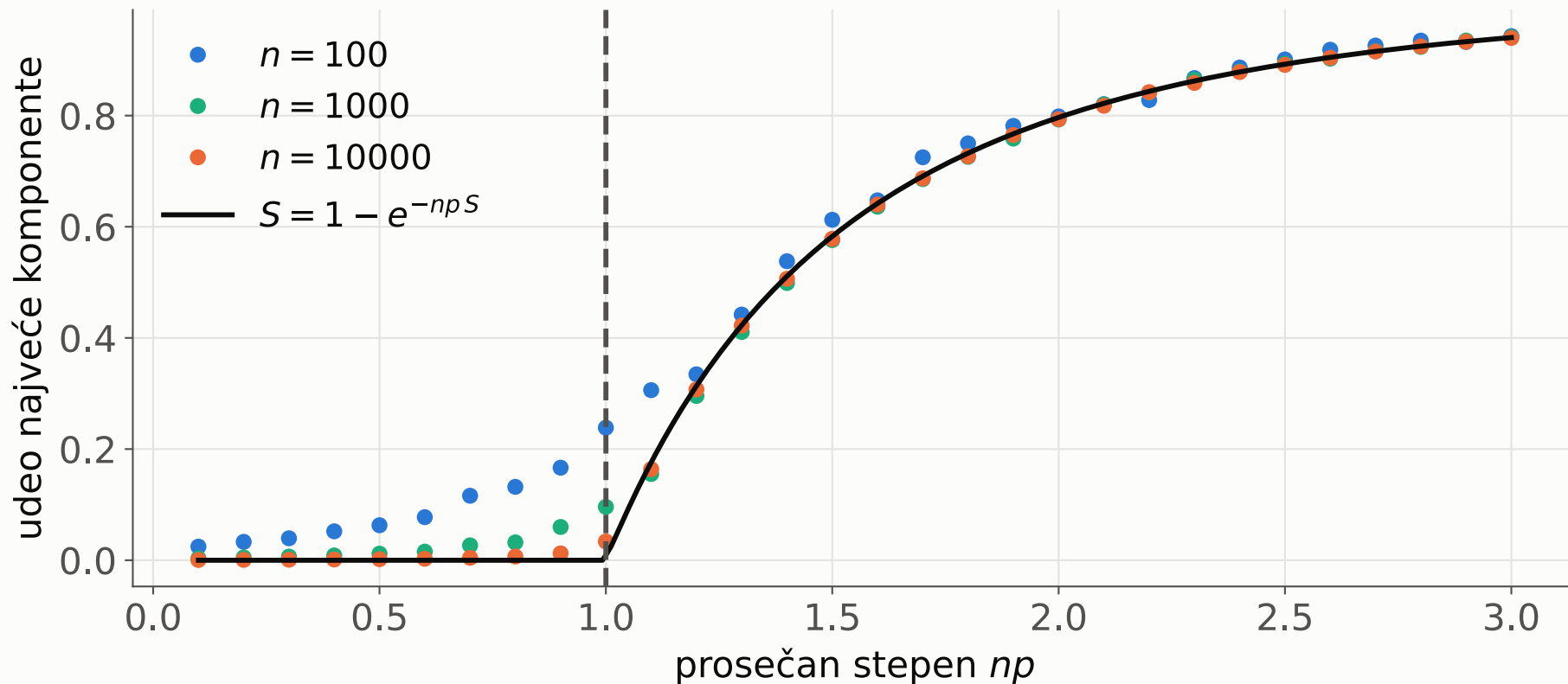
$$|E| \sim B\left(\binom{n}{2}, p\right)$$

Stepen čvora



$$\deg(v) \sim B(n - 1, p) \xrightarrow{n \rightarrow \infty, np = \lambda} \text{Poisson}(\lambda)$$

Hipoteza: džinovska komponenta

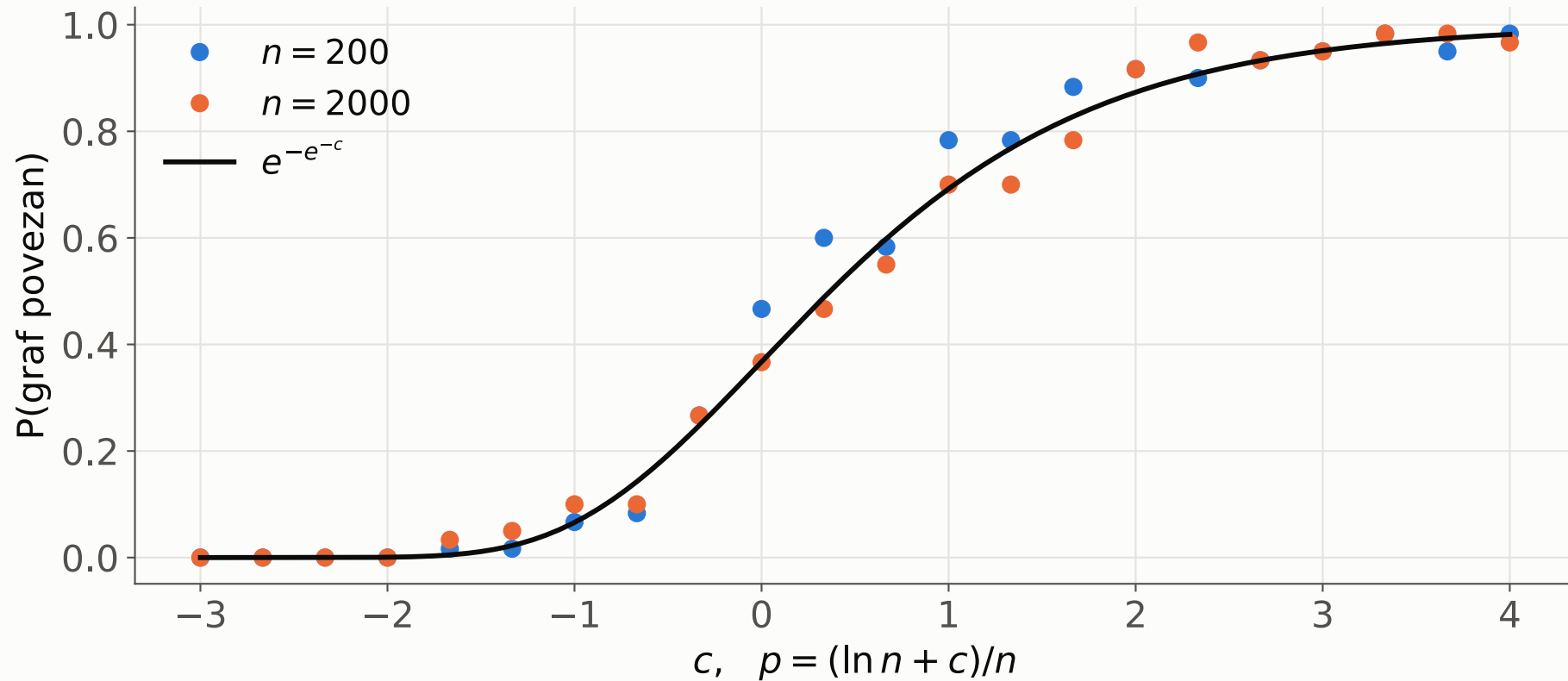


$$np < 1 : S \rightarrow 0$$

$$np > 1 : S = 1 - e^{-npS}$$

Fazni prelaz — oštar tek kad $n \rightarrow \infty$.

Kada je graf povezan?

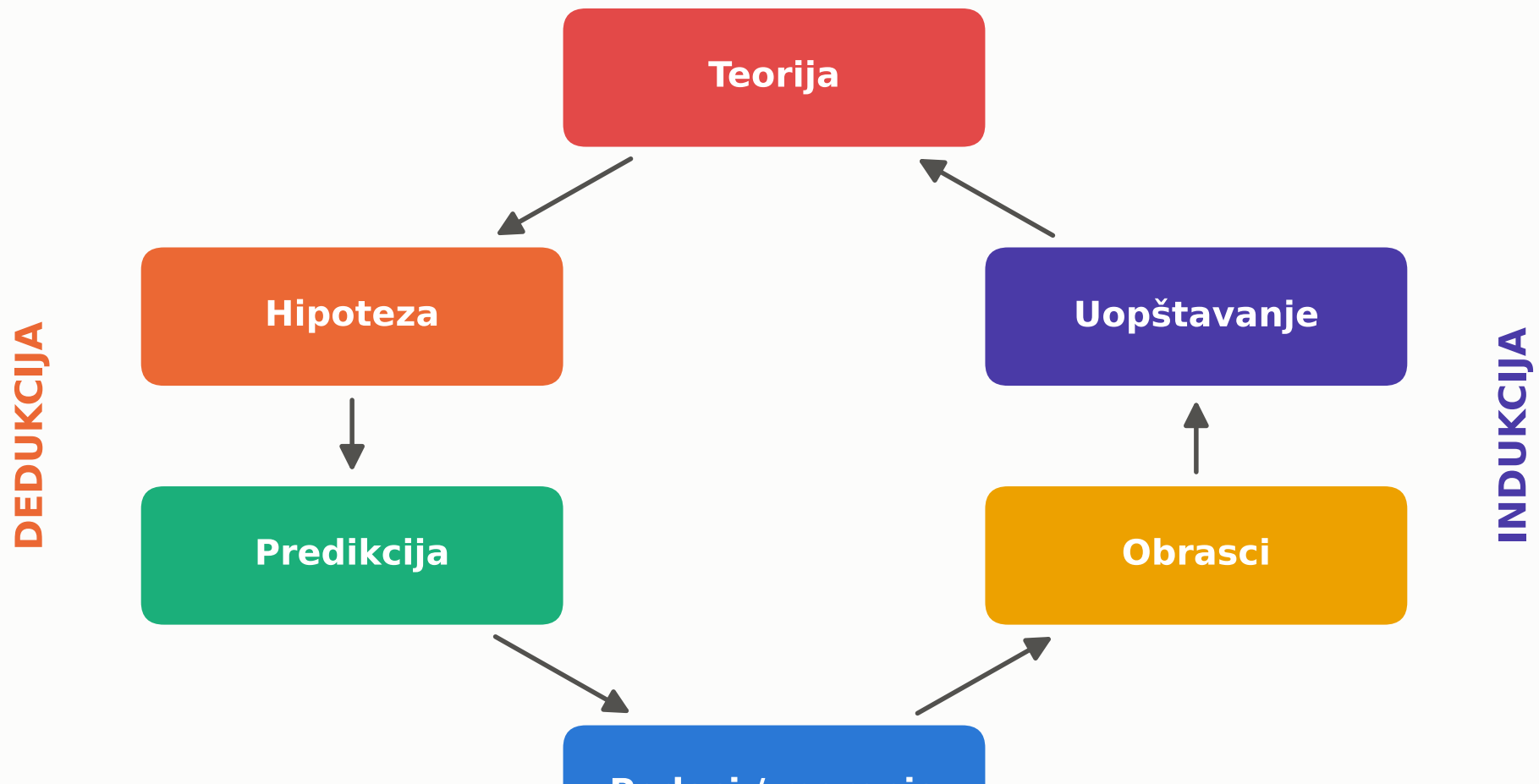


$$p = \frac{\ln n + c}{n} \implies P(G \text{ povezan}) \rightarrow e^{-e^{-c}}$$

VII

Od rezultata do teorije

Indukcija i dedukcija



Hipoteza → teorija

Korak	Pitanje
hipoteza	da li je proverljiva?
eksperiment	da li podaci slažu?
ponovljivost	da li drugi dobijaju isto?
objašnjenje	zašto važi? (model, dokaz)
teorija	da li predviđa nešto novo ?

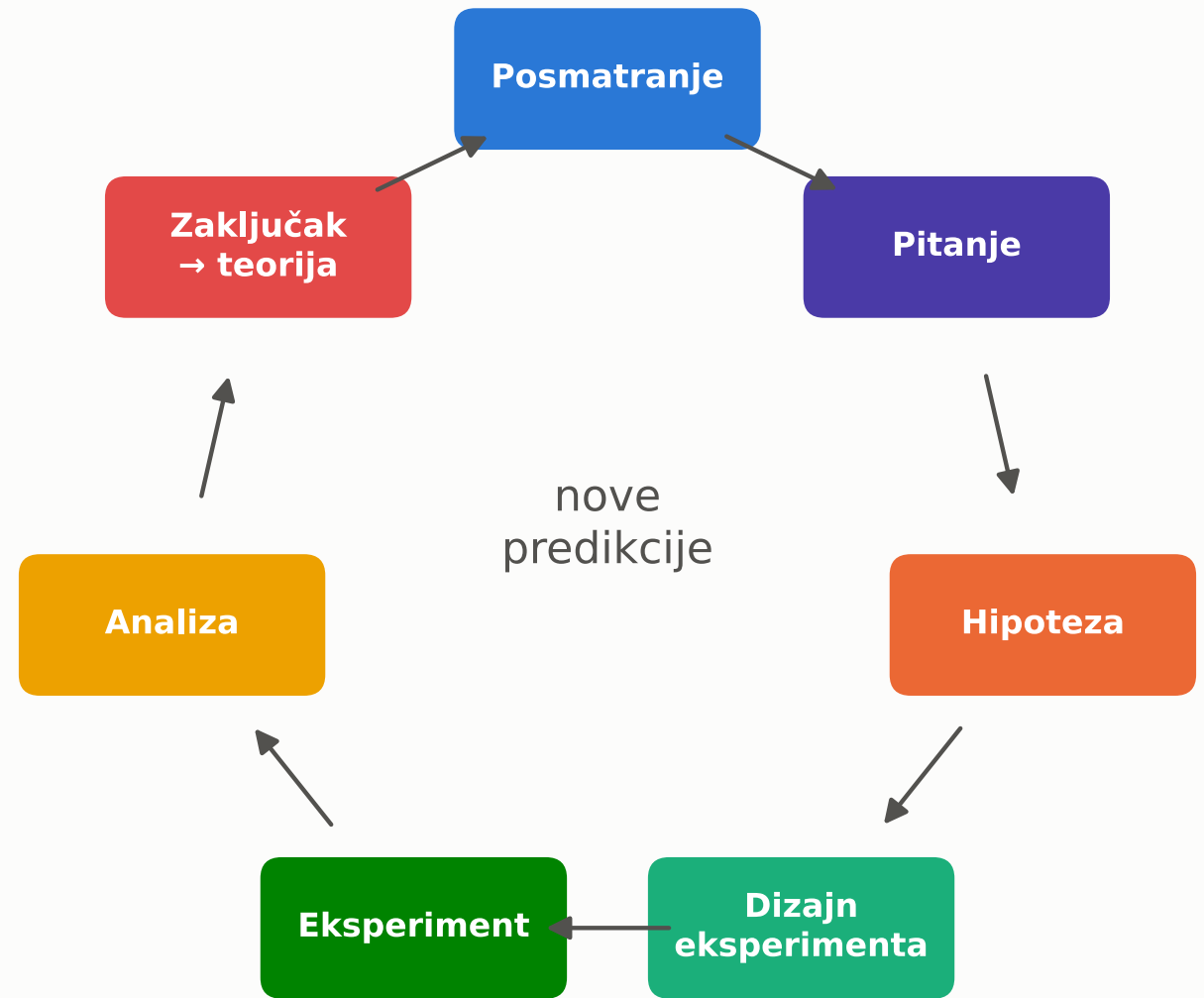
Šta smo videli

$$\underbrace{y = f(x) + \varepsilon}_{\text{eksperiment}} \longrightarrow \underbrace{\varepsilon \approx N(0, \sigma^2)}_{\text{CGT}} \longrightarrow \underbrace{\bar{x} \pm 1.96 \frac{s}{\sqrt{n}}}_{\text{odluka}}$$

$$\underbrace{C_n / (n \ln n)}_{\text{merenje}} \longleftrightarrow \underbrace{2(n+1)H_n - 4n}_{\text{teorija}}$$

I ispočetka

Svaka teorija je samo hipoteza koju **još nismo oborili**.



Pitanja?

Hvala!